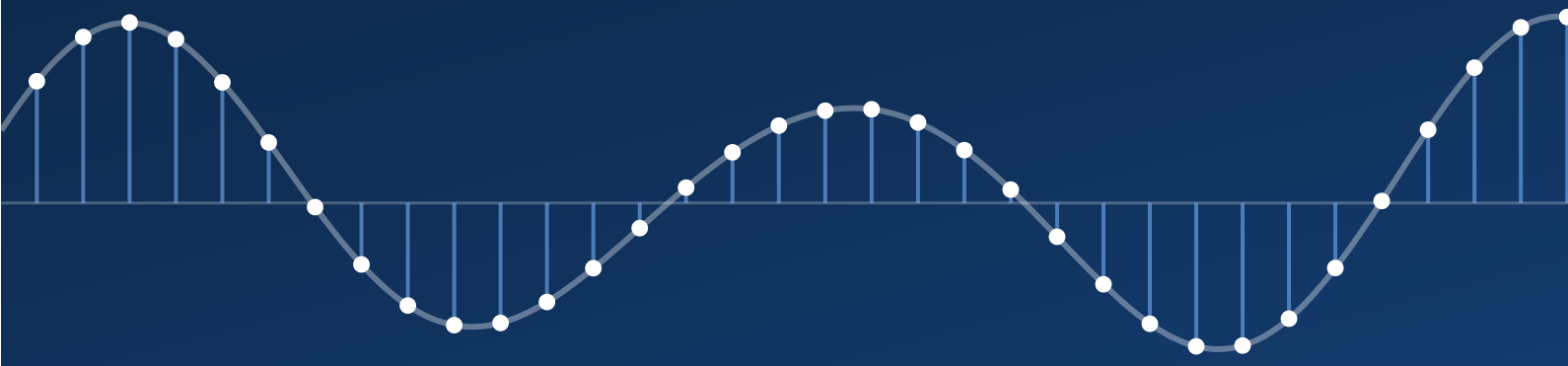
DIGITAL SIGNAL PROCESSING

REPORT EXPLAINED

شرح الريبورت سطر بسطر

نفس الشرح جملة بجملة في حجم أصغر: كل جملة ومعناها، واللي تقوله للدكتور، وأهم سؤال، والرسومات والجداول والمعادلات والمصطلحات.

Every sentence | Every term | Every figure & table | Every equation



295	55	14	32
جملة مشروحة	مصطلح	رسمة وجدول	معادلة

ELC 425 · Modern Academy

Line-by-line (compact)

المحتوى

إزاي تستخدم الملف

نفس الشرح جملة بجملة بس أصغر. كل فقرة: **(1)** كل جملة من الريبورت بالإنجليزي وتحتها معناها بالعربي؛ **(2)** اللي تقوله للدكتور: الفكرة ← السبب ← مثال/ربط؛ **(3)** أهم سؤال متوقع عليها. والرسومات والجداول والمعادلات والمصطلحات زي ما هم.

★ مراجعة التعليمات — Instructions check

I1 Part 1 content: types, uses, advantages, limitations, choosing, steps of design (in detail)

✓ Complete. Every item has its own section:

Instruction item	Where in your report
Types	1.5 Types of Digital Filters: FIR and IIR (p. 6)
Uses	1.2 Uses of Digital Filters (p. 4)
Advantages	1.3 (9 items, p. 5)
Limitations	1.4 (3 items, p. 6)
Choosing between them	1.6 + Example 5.1 + Table 1 (p. 7–10)
Steps of design *in detail*	1.7 + 1.7.1 → 1.7.5 + 1.8 examples (p. 10–19)

لو سألك «ليه الـ types جت بعد الـ limitations مع إن التعليمات بادئة بيها؟» قول: عشان الـ **choosing between them** لازم ييجي بعد الـ types على طول، فرتبت المقدمة الأول (uses, advantages, limitations) وبعدين الأنواع والاختيار بينهم.

I2 Page limits: Part 1 → 15–20 pages, Part 2 → 8–10 pages, total ≤ 30

✓ Part 1 = p. 4 → 19 = **16 pages** · Part 2 = p. 20 → 29 = **10 pages** · Total = **29 pages**.

⚠ الجزء التاني واقف على الحد بالظبط (10 صفحات شاملة الـ References)، فأي سطر زيادة هيعدّي الحد — شوف I3.

I3 Times New Roman, font 13, line spacing 1.5 — written in Word

⚠ **أهم نقطة تراجعها قبل المناقشة.** قست الـ PDF بتاعك: حجم الخط **13** ✓، بس الخط اللي جوه الـ PDF اسمه **Liberation Serif** (لأن الملف اتعمل على LibreOffice مش Word)، والمسافة بين السطور 18.6 pt ≈ — والـ 1.5 lines لخط 13 بتطلع حوالي 21.6–22.4 pt. يعني الـ spacing الظاهر ≈ **1.3 مش 1.5**.

اعمل إيه: افتح ملف الـ Word نفسه في Microsoft Word ← Ctrl+A ← بص على خانة الخط لازم تكون **Times New Roman** و **13** ← Paragraph ← Line spacing لازم تكون **1.5 lines**.

• لو لسه ينفع تعدّل وترفع: صلّح الاتنين. خلي بالك إن الـ 1.5 هتزوّد الصفحات حوالي 33–32 (≈ 15% صفحة) فهتعدّي الـ 30، والجزء التاني هيعدّي الـ 10. عشان ترجع تحت الحد شيل: Table 7 (هو نفس معلومات Fig. 5)، و Table 3 (بيكرر كلام 1.7.2)، وفقرة الـ Lattice الطويلة.

• لو مش هينفع تعدّل: ما تقولش للدكتور إنه 1.5 لو مش 1.5 — لو اتسألت، قول بصراحة إنك هتصلّحها.

I4 “Note: This page is the first page in your report.”

⚠ صفحة التعليمات عندك هي **صفحة 2** بعد الغلاف. الجملة بتقول صراحة إنها لازم تكون **أول صفحة** في الريبورت. ده أول حاجة ممكن يمسكها لأنه بيعمل check على التعليمات.

لو تقدر تعدّل: خلّي صفحة التعليمات **صفحة 1** وبعدها الغلاف. لو اتسألت وما عدّلتش: «I understood it as the first page after the cover» — بس الأأمن إنك تصلّحها.

I5 Cover page as given in the WhatsApp group

✓ Logo, branch, title "DSP Lecture Report – Fall 27 – Law 20", student table, supervisor, Cairo 2027 — all present.

⚠ فيه رقم «**1**» زيادة مكتوب في نص الغلاف تحت Cairo 2027، غير رقم الصفحة اللي تحت. امسحه، وقارن الغلاف بالنسخة اللي على الجروب حرف حرف (الألوان والخطوط).

I6 Upload two copies (Word + PDF) to Moodle — Thursday 22/10/2026, 8 PM

⚠ الميعاد عدّى (النهارده 24/10). اتأكد إن **النسختين** (.docx و .pdf) مرفوعين، لأنه بيقول two copies. صوّر صفحة الـ submission على Moodle عشان لو اتسألت.

I7 Other things he can check on the spot

✓ Table of contents page numbers match the pages ✓ · Figures numbered Fig. 1–5 with captions ✓ · Tables numbered 1–9 with titles ✓ · Every equation numbered ✓ · References section present ✓

⚠ **(1)** مفيش ولا citation جوه الكلام زي [1] أو [2] — الأحسن تضيف [1] بعد الفقرات المنقولة من الكتاب. **(2)** المعادلات في Part 1 مترقمة (5.7)...(5.1) يعني بترقيم Chapter 5 في الكتاب، وفي Part 2 مترقمة (2.23)...(2.1) — جهّز الإجابة: «Part 1 keeps the equation numbers of Chapter 5 of the reference so they can be checked against the book». **(3)** في Eq. (5.3) الـ $\Sigma\, h(k)x(n-k)$ مكتوب حدها العلوي N، والأصح ∞ زي (5.1) — خطأ كتابة صغير اعرفه. **(4)** المعادلة (5.7) مترقمة مرة واحدة والكلام بيقول (5.7a) و (5.7b). **(5)** المرجع [1] مكتوب ELC 422 — اتأكد إنه نفس الكود اللي على غلاف الكتاب.

I8 Attachment No. 1 — handwritten problems of Chapter 5

ده مش جزء من الريبورت: مسائل Chapter 5 تتحل **بخط الإيد على ورق A4 كويس** وتتسلّم لـ Eng. M. Khalaf يوم **التلاتاء 1/12/2026**. لو سألك عنها اعرف الميعاد.

★ The report in one minute — الريبورت في دقيقة

1 If he says: “Tell me about your report”

ده أول كلام تقوله — بيبيّن إنك عارف **الريبورت ماشي إزاي ومترتب ليه**، وده أكتر حاجة بتبيّن إنك اللي عامله.

My report has two parts. In Part 1, I explain digital filters: what they are, their uses, advantages and limitations, the two types FIR and IIR, how to choose between them, and the five design steps, with examples from Chapter 5. In Part 2, I explain the Laplace transform as the tool for continuous-time systems, then I go through sampling to discrete time and the z-transform, and I show the link between them, $z = e^{sT}$. The two parts are connected: Example 5.4 converts an analogue filter H(s) into a digital filter H(z), and Section 2.6 explains why this works.

2 My own decisions — why I built the report this way

الدكتور لما يسأل «ليه عملت كده؟» عايز يتأكد إنك صاحب القرار. دي قراراتك وسببها:

Decision	Why I did it
Order of Part 1	I put uses, advantages and limitations first, then the types, so that “choosing between FIR and IIR” comes right after the types.
Examples 5.1–5.5	Each example proves one point: 5.1 the cost of FIR vs IIR, 5.2 a specification, 5.3 the five steps together, 5.4 analogue → digital, 5.5 structures.
Example 5.4	I chose it because it is the bridge between my two parts — H(s) is Laplace, H(z) is the z-transform.
Sampling (2.4) in Part 2	Laplace is continuous and z is discrete, so I needed sampling in the middle to connect them.
Table 8 with Fs = 128 Hz	I used the sampling frequency of Example 5.3 so the Part 1 filter appears on the unit circle of Part 2.
Equation numbers 5.x	I kept the numbering of Chapter 5 of the reference so they can be checked against the book.
References	The course book and the lecture notes: Lectures 3, 4 (Chapter 1) and 9, 10-2, 11 (Chapter 4).

★ المصطلحات — Glossary

A–Z Every technical term in the report, in simple words

Term	معناه
Signal / analogue / digital	إشارة: كمية بتتغيّر مع الزمن. analog = ليها قيمة عند كل لحظة وأي قيمة. digital = عينات (أرقام) عند لحظات معيّنة وبعدد bits محدود.
Sample, x(n), n	عينة = قيمة الإشارة عند لحظة. x(n) = العينة رقم n. n = رقم العينة (0، 1، 2...).
Sampling period T / sampling frequency Fs	T = الزمن بين كل عينتين. Fs = 1/T = عدد العينات في الثانية.
Frequency f, ω, F, Ω	f بالـ Hz، ω = 2πf بالـ rad/s (متصل). F = f/Fs بالـ cycles/sample، ωT = Ω = 2πF بالـ rad/sample (متقطع).
Amplitude / phase	amplitude = حجم الموجة. phase = موقعها في الدورة (إزاحة زمنية).
Frequency response	إزاي الفلتر بيأثّر على كل تردد: magnitude (بيكبّر/يصغّر قد إيه) و phase (بيأخّر قد إيه).
Passband / stopband / transition band	passband = الترددات اللي بتعدّي. stopband = اللي بتتمنع. transition = المنطقة بينهم.
Lowpass / highpass / bandpass / bandstop	يعدّي الواطي / العالي / حتة في النص / يمنع حتة في النص.
δp, δs, fp, fs	أقصى انحراف مسموح في الـ passband والـ stopband، وحافة الـ passband وحافة الـ stopband.
dB (decibel)	20 log10 للنسبة. 60 dB = نسبة 1/1000.
Ripple / attenuation	ripple = تموّج في الـ passband. attenuation = قد إيه الإشارة بتقل في الـ stopband.
Noise	أي إشارة مش عايزينها راكبة على الإشارة الأصلية.
Impulse δ(n) / impulse response h(k)	δ(n) = 1 عند n=0 وصفر غير كده. h(k) = خرج النظام لما الدخل δ(n)؛ بيوصف النظام كله.
Convolution	$y(n) = \Sigma\, h(k)x(n-k)$: كل خرج = مجموع العينات الحالية والقديمة مضروبة في h(k).
Difference equation	معادلة بتحسب y(n) من عينات الدخل والخرج الحالية والقديمة — دي اللي بتتنفّذ فعلًا.
Coefficients (h(k), ak, bk)	الأرقام اللي بتحدد الفلتر. h(k) للـ FIR، ak (بسط) و bk (مقام/feedback) للـ IIR.
Recursive / non-recursive	recursive = بيستخدم الخرج القديم (feedback) = IIR. non-recursive = الدخل بس = FIR.
Feedback	جزء من الخرج بيرجع يدخل الحساب تاني.
FIR / IIR	Finite / Infinite Impulse Response: الـ h(k) محدود ولا مالوش آخر.
Order	أعلى أس لـ z^{-1} (في المقام للـ IIR) = عدد الـ delays الأساسية.
Transfer function H(s), H(z)	Y/X في دومين لابلاس أو Z. بسط ومقام.
Poles / zeros	pole: المقام = ∞). zero: البسط = 0 (H = 0).
Unit circle	دايرة |z| = 1 في الـ z-plane = محور الترددات الرقمي.
s-plane / z-plane	مستوى الأرقام المركّبة لـ s (d أفقي، jω رأسي) ولـ z (Re، Im).
ROC	Region Of Convergence: قيم z اللي الـ Σ عندها بتتجمع لرقم محدود.
Stable / BIBO	أي دخل محدود يدّي خرج محدود. شرطه $\Sigma\lvert h(k)\rvert < \infty$ = كل الـ poles جوه الدايرة.
Causal	الخرج مبيعتمدش على المستقبل.
LTI	Linear Time-Invariant: خطي (التجميع يتحفظ) وثابت مع الزمن (تأخير الدخل = تأخير الخرج).

Linear phase / phase distortion	linear phase = كل الترددات تتأخر نفس الزمن ← مفيش تشويه لشكل الموجة. phase distortion = ترددات تتأخر أكتر من غيرها ← الشكل يتشوّه.
Aliasing	تردد عالي بعد الـ sampling يبان كأنه تردد واطي (لما Fs ≤ 2fmax).
Nyquist rate	2fmax: أقل Fs يمنع الـ aliasing.
Quantization / round-off / overflow	تقريب لأقرب مستوى (ADC والـ coefficients) / تقريب نتيجة الحساب / الناتج أكبر من أكبر رقم ممكن.
Word length / fixed-point	عدد الـ bits لكل رقم / تمثيل الأرقام بنقطة عشرية ثابتة (أبسط وأسرع من floating).
Accumulator	register بيجمع فيه حواصل الضرب (في TMS32010 = 32 bit).
Structure / realization	شكل توصيل الـ adders والـ multipliers والـ delays اللي بينفّذ H(z).
Direct / cascade / parallel	H(z) كما هي / حاصل ضرب sections ورا بعض / مجموع sections جنب بعض.
Second-order section (biquad)	H(z) صغيرة بسطها ومقامها درجة تانية.
Transversal / tapped delay line	الـ FIR المباشر: سلسلة delays وكل نقطة تتضرب في h(k) وتتجمع.
Lattice	structure من stages متكررة كل واحدة بمعامل K.
z^{-1} block	تأخير عينة واحدة (memory).
Partial fractions	تفكيك كسر كبير لمجموع كسور صغيرة.
Impulse invariant / bilinear	طريقتين analog ← digital: الأولى تحافظ على h(t) وفيها aliasing، التانية تحافظ على الـ magnitude ومفيهاش aliasing.
Butterworth / Chebyshev / elliptic	أشكال فلاتر analog كلاسيكية: ناعم بدون ripple / أحدّ مع ripple في جنب / أحدّ واحد مع ripple في الجنبين.
Window / frequency sampling / optimal	طرق حساب coefficients الـ FIR.
Real-time / batch	لازم يخلص قبل العينة الجاية / الداتا متخزنة ومش مستعجلين.
ADC / DAC	Analogue-to-Digital / Digital-to-Analogue converter.
ROM / RAM / ALU	ذاكرة للقراية بس (coefficients) / ذاكرة قراية وكتابة (العينات) / وحدة الحساب.
DSP processor (TMS32010, TMS320C25, DSP56000)	معالجات مخصوصة للـ DSP فيها hardware multiplier بيضرب في cycle واحدة.
VLSI	تصنيع ملايين الترانزستورات في chip واحدة.
CAD	Computer-Aided Design: برامج التصميم بالكمبيوتر (MATLAB...).
FFT	Fast Fourier Transform: طريقة سريعة لحساب الـ DFT (الترددات).
Adaptive filter	فلتر بيغيّر الـ coefficients بتاعته أوتوماتيك حسب الإشارة.
Fourier series / harmonic	كتابة إشارة periodic كمجموع sinusoids ترددتها kf0. الـ harmonic رقم k = المركّبة بتردد kf0.
Laplace transform	$X(s) = \int x(t)e^{-st}dt$: بيحوّل الإشارة المتصلة لدالة في s.
Z-transform	$X(z) = \Sigma x(n)z^{-n}$: بيحوّل الإشارة المتقطعة لدالة في z.

★ مقدمة الفلاتر وأنواعها — Part 1 · 1.1 – 1.5

1.1-a "A filter is a system or network that selectively changes ..." p. 4

الريبورت بيقول إيه — جملة بجملة

1 A filter is a system or network that selectively changes the wave shape, the amplitude–frequency characteristic and/or the phase–frequency characteristic of a signal in a desired manner.

الفلتر = نظام بيغيّر الإشارة **بشكل انتقائي** (selectively) يعني مش بيغيّر كل حاجة بنفس الدرجة. بيغيّر 3 حاجات: **شكل الموجة**، والـ **amplitude عند كل تردد** (مين يعدّي ومين يتشال)، والـ **phase عند كل تردد** (كل تردد يتأخر قد إيه). و«in a desired manner» = بالطريقة اللي إحنا عايزينها.

2 The common filtering objectives are to improve the quality of a signal (for example, to remove or reduce noise), to extract information from signals, or to separate two or more signals that were previously combined, for example to make efficient use of an available communication channel.

ليه بنعمل filtering؟ 3 أهداف: (1) **نحسّن جودة الإشارة** = نشيل أو نقلّل الـ noise. (2) **نطلّع معلومة** من الإشارة (زي إننا نطلّع نبض القلب من إشارة ECG). (3) **نفصل إشارتين** كانوا متجمّعين، زي لما كذا إشارة بيتبعتوا على نفس الـ channel عشان نستغل الـ channel كويس، وفي الاستقبال الفلتر بيفصلهم.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	A filter changes a signal on purpose — it keeps the frequencies we want and weakens the ones we don't, or changes their phase.
WHY / HOW	Every signal is a sum of frequencies, and noise usually sits in frequencies different from the useful signal, so treating each frequency differently is how we clean or separate signals.
EXAMPLE · LINK	For example, an ECG signal is a few hertz while mains hum is 50 Hz, so a lowpass filter removes the hum and keeps the heart signal.

الأسئلة المتوقعة

1 What are the three filtering objectives?

Improve the quality of a signal (remove/reduce noise), extract information from signals, and separate two or more combined signals.

1.1-b "A digital filter is a mathematical algorithm, implemented in hardware and/or software ..." p. 4

الريبورت بيقول إيه — جملة بجملة

1 A digital filter is a mathematical algorithm, implemented in hardware and/or software, that operates on a digital input signal to produce a digital output signal in order to achieve a filtering objective.

الـ digital filter = **algorithm رياضي** (خطوات حسابية / معادلة). ممكن يتنفّذ **hardware** (دايرة رقمية / DSP chip) أو **software** (برنامج). بياخد إشارة **رقمية** ويطلّع إشارة **رقمية**، والهدف واحد من أهداف الـ filtering اللي فاتت.

2 The term digital filter refers to the specific hardware or software routine that performs the filtering algorithm.

يعني لما نقول «digital filter» إحنا بنقصد الحاجة الفعلية اللي بتنفّذ الـ algorithm — سواء الـ hardware أو الـ routine (البرنامج).

3 Digital filters often operate on digitized analogue signals, or just on numbers, representing some variable, that are stored in a computer memory.

الفلتر الرقمي بيشتغل على حاجتين: (1) إشارة analog اتعملها **digitization** (اتحوّلت أرقام بالـ ADC)، أو (2) **أرقام متخزنة** في الـ memory بتمثّل أي متغيّر (زي قراءات حرارة متسجّلة) — يعني مش لازم يكون فيه إشارة analog أصلًا.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	A digital filter is not a circuit — it is an equation that the processor applies to numbers.
WHY / HOW	Once the signal is a sequence of numbers x(n), filtering becomes just multiplying and adding samples according to the difference equation, so it can run in hardware or as a program.
EXAMPLE · LINK	That is why the same digital filter can work on live samples from an ADC or on data already stored in memory.

الأسئلة المتوقعة

1 What is the difference between an analogue and a digital filter?

An analogue filter is a circuit (R, L, C, op-amps) working on continuous signals; a digital filter is an algorithm working on number sequences x(n) → y(n).

1.1-c Fig. 1 — real-time digital filter with analogue input and output p. 4

الريبورت بيقول إيه — جملة بجملة

1 Fig. 1 shows a simplified block diagram of a real-time digital filter with analogue input and output signals.

Fig. 1 = رسمة مبسّطة لفلتر رقمي شغّال **real-time** (بيطلّع الخرج أول بأول مع الدخل) والدخل والخرج بتوعه **analog**.

2 The band-limited analogue signal is sampled periodically and converted into a series of digital samples x(n), n = 0, 1, ….

band-limited = الإشارة متحدّدة الترددات (مفيهاش ترددات أعلى من حد معيّن) — ده شرط عشان نمنع الـ aliasing.
بعدين بناخد منها عينات **كل فترة ثابتة** (periodically) ونحوّلها لأرقام x(0), x(1), x(2)… — و n هو رقم العينة.

3 The digital processor implements the filtering operation by mapping the input sequence x(n) into the output sequence y(n) in accordance with the computational algorithm of the filter.

الـ processor هو الفلتر نفسه: بياخد الأرقام x(n) ويحسب منها أرقام جديدة y(n) باستخدام **معادلة الفلتر** (الـ difference equation). «mapping» = تحويل sequence لـ sequence.

4 The DAC converts the digitally filtered output into analogue values, which are then analogue filtered to smooth them and to remove unwanted high-frequency components.

الـ DAC بيرجّع الأرقام y(n) لقيم analog، بس الخرج بيطلع على شكل **سلالم** (كل قيمة ثابتة لحد العينة اللي بعدها). عشان كده في الآخر فيه **analogue filter** (lowpass) بينعّم السلالم دي ويشيل الترددات العالية اللي جت من الحواف الحادة.

اشرحها كده عشان يبان إنك فاهم

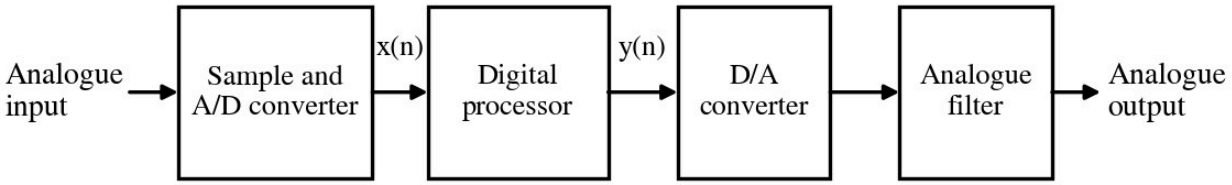
THE IDEA	Fig. 1 shows the full real-time chain: analogue in → ADC → processor → DAC → analogue filter → analogue out.
WHY / HOW	The input is band-limited first to prevent aliasing, and the output analogue filter is needed because the DAC gives a staircase signal with high-frequency steps.
EXAMPLE · LINK	So the digital filter itself is only the middle block; the blocks around it are what let it work on real analogue signals.

الأسئلة المتوقعة

1 Why is the input band-limited before sampling?

To avoid aliasing — the signal must have no components above Fs/2 (sampling theorem, Eq. 2.11).

Fig. 1 Block diagram of a real-time digital filter p. 4



الرسمة / الجدول فيه إيه — جزء جزء

1. **Analogue input**: الإشارة الحقيقية (صوت، ECG...) — متصلة في الزمن.
2. **Sample and A/D converter**: بياخد عينات كل T ثانية ويحوّل كل عينة لرقم بعدد bits محدود (+ sampling + quantization + coding) ← يطلع **x(n)**.
3. **x(n)** على السهم: يعني من هنا ورايح الإشارة بقت أرقام (sequence).
4. **Digital processor**: ده الفلتر نفسه — بينفّذ الـ difference equation على x(n) ويطلّع **y(n)**.
5. **D/A converter**: يرجّع y(n) لقيم analog، بس الخرج بيبقى سلالم (staircase).
6. **Analogue filter**: lowpass في الآخر ينعّم السلالم ويشيل الترددات العالية ← **Analogue output**.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Fig. 1 shows that a digital filter can process analogue signals by placing it between an ADC and a DAC.
WHY / HOW	The digital part can only handle numbers, so the signal must be converted to x(n) before the processor and back to analogue after it; the input must be band-limited to Fs/2 to avoid aliasing, and the output filter smooths the DAC steps.
EXAMPLE · LINK	This same chain appears in Part 2: the A/D block is the sampling of Section 2.4, $x(n) = x_a(nT)$.

الأسئلة المتوقعة

1 Which block is the digital filter itself?

The digital processor — the other blocks only convert the signal to and from numbers.

1.2 Uses of Digital Filters p. 4–5

الريبورت بيقول إيه — جملة بجملة

1 The filtering objectives given in Section 1.1 (noise reduction, extraction of information and separation of combined signals) define what filters are used for.

الاستخدامات أساسًا هي نفس الأهداف التلاتة اللي في 1.1: نشيل noise، نطلّع معلومة، نفصل إشارات.

2 Digital filters play very important roles in DSP, and compared with analogue filters they are preferred in a number of applications, because of the advantages listed in Section 1.3.

الفلاتر الرقمية مهمة جدًا في الـ DSP، وفي تطبيقات كتير بنفضّلها على الـ analog — والسبب هو المزايا اللي في 1.3 (هنشرحها).

3 The applications mentioned in the course reference are: data compression; biomedical signal processing; speech processing; image processing; data transmission; digital audio; telephone echo cancellation.

التطبيقات من الكتاب: **data compression** (ضغط البيانات — الفلتر يقسم الإشارة لترددات ونشيل الأقل أهمية)، **biomedical** (ECG, EEG)، **speech** (تحسين الصوت / التعرّف عليه)، **image** (تنعيم / توضيح الصور)، **data transmission** (المودم والاتصالات)، **digital audio** (equalizers)، **echo cancellation** في التليفون (شيل الصدى اللي بيرجع للمتكلّم).

4 Several further uses follow from the nature of digital filters.

وفيه استخدامات زيادة جاية من **طبيعة** الفلتر الرقمي نفسه — دي إضافتك إنت على قائمة الكتاب.

5 Because the frequency response of a digital filter can be adjusted automatically when it is implemented on a programmable processor, digital filters are widely used as adaptive filters.

adaptive filter = فلتر بيغيّر نفسه لوحده حسب الإشارة (زي echo canceller بيتعلّم شكل الصدى). ده ممكن في الرقمي لأن الـ frequency response بيتحدد بأرقام (coefficients)، والـ processor يقدر يغيّرها أوتوماتيك وهو شغّال.

6 Digital filters can be used at very low frequencies, which are found in many biomedical applications and where analogue filters are impractical.

الإشارات الطبية (زي ECG) ترددااتها واطية جدًا (أقل من Hz لعشرات الـ Hz). الفلتر الـ analog للترددات دي محتاج capacitors و inductors ضخمة جدًا فمش عملي، أما الرقمي فمش فارقة معاه.

7 In addition, filters are sometimes used to equalize or compensate the phase response of a system, or as phase shifters (Section 1.7.1).

كمان الفلتر ممكن يتستخدم عشان **يصلّح الـ phase** بتاع نظام تاني (equalizer: لو قناة بتأخّر ترددات أكتر من غيرها، الفلتر يعوّض ده)، أو كـ **phase shifter** (يزحزح الـ phase من غير ما يغيّر الـ amplitude).

8 A practical use is given in Example 5.3 (Section 1.8), where a digital filter is designed for real-time physiological noise reduction in a biomedical signal processing project.

ومثال عملي في الريبورت: Example 5.3 — فلتر real-time بيشيل الـ noise من إشارة فسيولوجية (من جسم الإنسان).

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The uses come from the three filtering objectives: removing noise, extracting information and separating signals.
WHY / HOW	Digital filters are preferred in these applications because of their advantages — they can be reprogrammed, they work at very low frequencies and they can have linear phase.
EXAMPLE · LINK	Biomedical filtering is the clearest case: the signals are at a few hertz, where analogue filters are impractical, which is exactly Example 5.3 in my report.

الأسئلة المتوقعة

1 **Why are digital filters used as adaptive filters?**

Because on a programmable processor the frequency response can be changed automatically, just by changing the coefficients.

1.3 Advantages of Digital Filters (9 points) p. 5

الريبورت بيقول إيه — جملة بجملة

1 Compared with analogue filters, digital filters have one or more of the following advantages:

المقارنة دايمًا **بالـ analog**: الرقمي عنده واحدة أو أكتر من المزايا دي.

2 1. Characteristics not possible with analogue filters. A digital filter can have, for example, a truly linear phase response.

حاجات الـ analog ما يقدرش يعملها، أهمها **linear phase بالظبط**: يعني كل الترددات تتأخر **نفس الزمن** فشكل الموجة ما يتشوّهش. الـ analog بيقرّب بس عمره ما يوصل بالظبط.

3 2. Stable performance. The performance of a digital filter does not vary with environmental changes such as thermal variations, which eliminates the need to calibrate the filter periodically.

الأداء **ثابت**: قيم المقاومات والمكثفات في الـ analog بتتغيّر مع الحرارة والزمن فلازم نعمل calibration كل فترة، أما الرقمي فهو أرقام في الـ memory مبتتغيّرش.

4 3. Programmability. When a programmable processor is used, the frequency response can be adjusted automatically, which is why digital filters are widely used as adaptive filters.

قابل للبرمجة: تغيّر الـ coefficients ← يتغيّر الـ response، من غير ما تغيّر ولا قطعة — وعشان كده بيتستخدم adaptive.

5 4. Shared hardware. Several input signals or channels can be filtered by one digital filter without the need to replicate the hardware.

فلتر رقمي **واحد** يقدر يفلتر كذا channel (بالتبادل، time-sharing) — في الـ analog كل channel محتاجة دايرة لوحدها.

6 5. Data storage. Both filtered and unfiltered data can be saved for further use.

تقدر **تخزّن** الداتا قبل الفلترة وبعدها وترجعلها بعدين.

7 6. VLSI technology. The advances in VLSI technology can be used to fabricate digital filters that are small in size, consume low power and keep the cost down.

VLSI = Very Large Scale Integration (ملايين الترانزستورات في chip واحدة) ← الفلتر يبقى صغير، استهلاكه قليل، ورخيص.

8 7. High precision. The precision achievable with analogue filters is restricted; typically only about 60 to 70 dB of stopband attenuation is possible with active filters designed with off-the-shelf components. With digital filters the precision is limited only by the word length used.

دقة عالية: الـ analog الـ active (بـ op-amps وقطع عادية من السوق = off-the-shelf) أقصاه يقلّل الـ stopband بحوالي 60–70 dB بسبب tolerance القطع. الرقمي الدقة بتاعته محدودة بس بعدد الـ bits (word length) — تزوّد bits تزيد الدقة.

9 8. Repeatability. The performance of digital filters is repeatable from unit to unit.

لو صنعت 1000 نسخة من نفس الفلتر الرقمي، كلهم **نفس الأداء بالظبط** — في الـ analog كل نسخة بتفرق شوية بسبب اختلاف القطع.

10 9. Very low frequencies and wide frequency range. Digital filters can be used at very low frequencies … They can also be made to work over a wide range of frequencies by a mere change of the sampling frequency.

يشتغل في **الترددات الواطية جدًا**، وكمان تقدر تنقله لـ range ترددات تاني **بمجرد تغيير الـ Fs** — لأن الفلتر الرقمي مصمّم على F = f/Fs، فلو غيّرت Fs كل الترددات بتاعته بتتنقل معاها.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	All nine advantages come from one fact: a digital filter is defined by numbers, not by component values.
WHY / HOW	Numbers don't drift with temperature or age, so the filter is stable and repeatable; numbers can be changed, so it is programmable; and precision depends only on how many bits we use.
EXAMPLE · LINK	For example, an active analogue filter stops at about 60–70 dB of attenuation because of component tolerances, while a digital filter can go further just by using a longer word length.

الأسئلة المتوقعة

1 Why don’t digital filters need periodic calibration?

Their performance does not vary with environmental changes such as thermal variations — the coefficients are numbers, not component values.

1.4 Limitations of Digital Filters (3 points) p. 6

الريبورت بيقول إيه — جملة بجملة

1 The main disadvantages of digital filters compared with analogue filters are the following:

العيوب برضه بالمقارنة بالـ analog — 3 عيوب.

2 1. Speed limitation. The maximum bandwidth of the signals that digital filters can handle in real time is much lower than for analogue filters.

السرعة: أقصى bandwidth يقدر الرقمي يتعامل معاه real-time أقل بكتير من الـ analog (الـ analog بيشتغل لحد GHz، الرقمي محدود بسرعة التحويل والحساب).

3 The conversion time of the ADC and the settling time of the DAC limit the highest frequency that can be processed.

conversion time = الوقت اللي الـ ADC بياخده عشان يحوّل عينة. **settling time** = الوقت اللي الـ DAC بياخده لحد ما خرجه يستقر. لو دول بطيئين ← مش هنقدر نزوّد Fs ← أعلى تردد (Fs/2) يبقى محدود.

4 In addition, the speed of operation depends on the speed of the digital processor and on the number of arithmetic operations required by the filtering algorithm, and this number increases as the filter response is made tighter.

كمان كل عينة محتاجة حسابات (ضرب وجمع) لازم تخلص قبل العينة اللي بعدها. كل ما الـ response يبقى **أضيق/أحدّ** (tighter) نحتاج coefficients أكتر ← حسابات أكتر ← أبطأ.

5 2. Finite word length effects. Digital filters are subject to ADC noise resulting from quantizing a continuous signal, and to round-off noise incurred during computation.

محدودية عدد الـ bits: (1) الـ ADC بيقرّب كل عينة لأقرب مستوى (quantization) ← فيه خطأ = noise. (2) أثناء الحساب نتيجة الضرب محتاجة bits أكتر فبنقرّبها ← noise (round-off) تاني.

6 With higher-order recursive filters, the accumulation of round-off noise could lead to instability.

في الفلاتر **الـ recursive** (IIR) الخرج بيرجع يدخل تاني (feedback)، فالخطأ بيرجع ويتراكم — ولو الـ order عالي ممكن الفلتر يبقى **unstable**.

7 3. Long design and development times. The design and development times, especially for hardware, can be much longer than for analogue filters.

وقت التصميم طويل، خصوصًا لو هنعمل hardware مخصوص.

8 However, once developed, the hardware and/or software can be used for other filtering or DSP tasks with little or no modification.

بس الميزة إن بعد ما يخلص، نفس الـ hardware/software يتستخدم لمهام تانية بتعديل بسيط (نغيّر الـ coefficients أو البرنامج).

9 Good computer-aided design (CAD) support can make the design of digital filters an enjoyable task, but some expertise is required to use such design aids effectively.

برامج الـ **CAD** (زي MATLAB) بتسهّل التصميم جدًا، بس محتاج خبرة عشان تستخدمها صح.

اشرحها كده عشان يبان إنك فاهم

THE IDEA The limitations are the price of working with numbers: speed, finite bits, and design effort.

WHY / HOW Every sample has to be converted and computed before the next one arrives, so the ADC, DAC and processor speed limit the bandwidth; and since numbers are stored with limited bits, quantization and round-off noise appear.

EXAMPLE · LINK This is linked to Section 1.7.4: in recursive filters the round-off noise is fed back, which is why it can even cause instability.

الأسئلة المتوقعة

1 What limits the maximum frequency a digital filter can handle?

The ADC conversion time, the DAC settling time, the processor speed and the number of arithmetic operations of the algorithm.

1.5-a "Digital filters are broadly divided into two classes ..." — Eqs. (5.1), (5.2) p. 6

$$y(n) = \Sigma_{k=0}^{\infty} h(k)\, x(n-k) \quad (5.1, \text{IIR}) \quad | \quad y(n) = \Sigma_{k=0}^{N-1} h(k)\, x(n-k) \quad (5.2, \text{FIR})$$

الريبورت بيقول إيه — جملة بجملة

1 Digital filters are broadly divided into two classes: infinite impulse response (IIR) filters and finite impulse response (FIR) filters.

نوعين بس: **IIR** = الـ impulse response بتاعه **لا نهائي**، و**FIR** = الـ impulse response **محدود**.

2 In its basic form, either type can be represented by its impulse response sequence h(k), k = 0, 1, ….

impulse response h(k) = خرج الفلتر لما الدخل يكون نبضة واحدة 1) δ(n) عند n=0 وصفر غير كده). الـ h(k) بتوصف الفلتر بالكامل.

3 The input and output signals of the filter are related by the convolution sum, given in Eq. (5.1) for the IIR filter and in Eq. (5.2) for the FIR filter.

convolution = العلاقة بين الدخل والخرج: $y(n) = \Sigma\, h(k)x(n-k)$. يعني كل خرج = مجموع العينات الحالية والقديمة مضروبة في أوزان h(k). (5.1) للـ IIR (الـ Σ لحد ∞) و (5.2) للـ FIR (لحد N−1).

4 It is evident from these equations that the impulse response of an IIR filter is of infinite duration, whereas that of an FIR filter is of finite duration, since h(k) has only N values.

من المعادلتين واضح: الـ IIR الـ h(k) بتاعه مالهاش آخر، والـ FIR ليها **N قيمة بس** (h(0) لحد h(N−1)).

اشرحها كده عشان يبان إنك فاهم

THE IDEA Both filter types can be written as a convolution of the input with the impulse response h(k); the only difference is the length of h(k).

WHY / HOW h(k) is how the filter responds to a single impulse. If it dies after N samples, the filter is FIR; if it continues forever, the filter is IIR.

EXAMPLE · LINK So Eq. 5.1 and 5.2 are the same equation — the sum goes to infinity for IIR and stops at N−1 for FIR.

الأسئلة المتوقعة

1 What does “impulse response” mean?

h(k) is the output of the filter when the input is a unit impulse δ(n).

1.5-b “In practice it is not feasible to compute the output of an IIR filter with Eq. (5.1) ...” — Eq. (5.3) p. 7

$$y(n) = \Sigma_{k=0}^{N} a_k\, x(n-k) - \Sigma_{k=1}^{M} b_k\, y(n-k) \quad (5.3)$$

الريبورت بيقول إيه — جملة بجملة

1 In practice it is not feasible to compute the output of an IIR filter with Eq. (5.1), because the impulse response is too long (infinite in theory).

مينفعش نحسب IIR بـ (5.1) لأننا هنحتاج نجمع عدد لا نهائي من الحدود ونخزّن عدد لا نهائي من العينات — مستحيل عمليًا.

2 Instead, the IIR filtering equation is expressed in a recursive form, Eq. (5.3), where ak and bk are the coefficients of the filter.

الحل: نكتبها **recursive** بنستخدم (5.3): $y(n) = \Sigma\, a_k x(n-k) - \Sigma\, b_k y(n-k)$. **الخروج القديمة** $y(n-k)$ بدل ما نجمع للأبد. a_k و b_k = معاملات الفلتر.

3 Equations (5.2) and (5.3) are therefore the difference equations of the FIR and IIR filters respectively.

difference equation = معادلة بتحسب الخرج من عينات (حالية ومتأخرة). (5.2) بتاعة الـ FIR و (5.3) بتاعة الـ IIR.

4 These equations, and in particular the values of h(k) for an FIR filter or of ak and bk for an IIR filter, are often the main objectives of a filter design problem.

هدف التصميم كله في الآخر = نلاقي **الأرقام**: $h(k)$ للـ FIR أو a_k و b_k للـ IIR.

5 In Eq. (5.3) the current output sample y(n) is a function of past outputs as well as present and past input samples; in other words, the IIR filter is a feedback system of some sort.

في الـ IIR الخرج الحالي بيعتمد على **الخرج القديم** + الدخل الحالي والقديم ← يعني فيه **feedback** (الخرج بيرجع يدخل).

6 In the FIR equation, by contrast, y(n) depends only on the present and past values of the input.

أما الـ FIR فالخرج بيعتمد على **الدخل بس** — مفيش feedback.

7 When all bk are set to zero, Eq. (5.3) reduces to the FIR Eq. (5.2).

لو b_k كلها = صفر يختفي جزء الـ feedback وتبقى (5.3) هي نفسها FIR — يعني الـ FIR حالة خاصة من المعادلة العامة.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	We cannot compute an infinite sum, so the IIR filter is computed recursively, using its own previous outputs.
WHY / HOW	Feeding back past outputs $y(n-k)$ through the b_k coefficients reproduces an infinitely long impulse response with only a few coefficients — that is what makes IIR efficient.
EXAMPLE · LINK	If all b_k are zero there is no feedback and Eq. 5.3 becomes the FIR equation 5.2, which shows FIR is a special case of the general form.

الأسئلة المتوقعة

1 Why is the IIR filter called recursive?

Its output y(n) is computed from past outputs y(n−k) — there is feedback.

1.5-c "Alternative representations ... are their transfer functions, Eqs. (5.4a) and (5.4b)"

p. 7

$$H(z) = \Sigma_{k=0}^{N-1} h(k)\, z^{-k} \quad (5.4a) \quad | \quad H(z) = \Sigma\, a_k z^{-k} / (1 + \Sigma\, b_k z^{-k}) \quad (5.4b)$$

الريبورت بيقول إيه — جملة بجملة

1 Alternative representations of the FIR and IIR filters are their transfer functions, Eqs. (5.4a) and (5.4b), which are very useful in evaluating the frequency responses of the filters.

طريقة تانية نوصف بيها الفلتر: **H(z)** = Y(z)/X(z) (transfer function). بنجيبها بالـ z-transform للـ difference equation. FIR: $H(z) = \Sigma h(k) z^{-k}$ (بسط بس). IIR: بسط (a_k) على مقام ($1 + \Sigma b_k z^{-k}$). فايدتها: نعوّض $z = e^{j\omega T}$ فيطلع الـ **frequency response** على طول.

2 The factors that influence the choices open to the designer at each stage of the design process are strongly linked to whether the filter is IIR or FIR.

كل قرار في التصميم (طريقة الحساب، الـ structure، عدد الـ bits) بيعتمد على الفلتر FIR ولا IIR.

3 It is therefore very important to appreciate the differences between the two types, their peculiar characteristics and, more importantly, how to choose between them.

عشان كده لازم نفهم الفرق بينهم وإزاي نختار — وده اللي في 1.6 (جملة ربط للسكشن اللي بعده).

اشرحها كده عشان يبان إنك فاهم

THE IDEA The transfer function H(z) is the same filter seen in the z-domain instead of the time domain.

WHY / HOW Using the delay property, every x(n−k) becomes $z^{-k}X(z)$, so the difference equation turns into a ratio of polynomials — and putting $z = e^{j\omega T}$ gives the frequency response directly.

EXAMPLE · LINK For FIR, H(z) has only a numerator, so all its poles are at z = 0 — this is why FIR is always stable, as I mention in 1.6.

الأسئلة المتوقعة

1 How do you get H(z) from the difference equation?

Take the z-transform using the delay property $x(n-k) \leftrightarrow z^{-k}X(z)$, then H(z) = Y(z)/X(z) (Eq. 2.18).

★ الاختيار بين FIR و IIR — Part 1 · 1.6

1.6-a The six points of comparison p. 7–8

الريبورت بيقول إيه — جملة بجملة

1 The choice between FIR and IIR filters depends largely on the relative advantages of the two filter types:

الاختيار بيعتمد على مزايا كل نوع مقارنة بالتاني — 6 نقط:

2 1. Phase response. FIR filters can have an exactly linear phase response, which means that no phase distortion is introduced into the signal by the filter.

linear phase = الـ phase بيتغيّر خطيًا مع التردد ← كل الترددات تتأخر **نفس الزمن** ← الموجة تطلع بنفس شكلها بس متأخرة = **مفيش phase distortion**. الـ FIR يقدر يوصل لده بالظبط (لو الـ h(k) متماثلة).

3 This is an important requirement in many applications, for example data transmission, biomedicine, digital audio and image processing.

ليه مهم؟ في data transmission تشوّه الشكل يعمل أخطاء في الـ bits، في الطب تشوّه شكل ECG يغيّر التشخيص، في الصوت والصورة بيبان.

4 The phase responses of IIR filters are nonlinear, especially at the band edges.

الـ IIR الـ phase بتاعه **مش خطي**، وأسوأ حاجة عند **حواف الـ band** (قرب الـ cutoff).

5 2. Stability. FIR filters realized non-recursively, that is by direct evaluation of Eq. (5.2), are always stable. The stability of IIR filters cannot always be guaranteed.

الـ FIR (لو اتنفّذ من غير feedback) **دايمًا stable** لأن خرجه مجموع محدود لدخل محدود. الـ IIR فيه feedback ← ممكن الخرج يكبر للأبد ← مش مضمون.

6 3. Finite word length. The effects of using a limited number of bits, such as round-off noise and coefficient quantization errors, are much less severe in FIR filters than in IIR filters.

أخطاء عدد الـ bits (round-off) وتقريب الـ coefficients أقل بكتير في الـ FIR — لأن مفيش feedback يرجّع الخطأ ويكبّره.

7 4. Number of coefficients. FIR filters require more coefficients than IIR filters for sharp cutoff responses. For a given amplitude response specification, more processing time and storage are therefore required for an FIR implementation.

عشان cutoff **حاد** الـ FIR محتاج coefficients أكتر بكتير ← وقت حساب وذاكرة أكتر (مثال 12 :5.1 مقابل 5).

8 However, one can readily take advantage of the computational speed of the FFT and of multi-rate techniques to improve the efficiency of FIR implementations significantly.

بس ممكن نعوّض ده بالـ **FFT** (fast convolution) و**multi-rate** (تغيير الـ sampling rate جوه النظام) فنقلّل الحسابات.

9 5. Analogue counterpart. Analogue filters can be readily transformed into equivalent IIR digital filters meeting similar specifications. This is not possible with FIR filters, as they have no analogue counterpart.

الـ IIR ممكن نجيبه بسهولة من فلتر **analog معروف** (Butterworth...) لأن الاتنين ليهم poles. الـ FIR مالوش «نظير» analog.

10 However, it is easier to synthesize FIR filters of arbitrary frequency responses.

بس لو عايز frequency response **بأي شكل غريب** (مش lowpass/highpass عادي) الـ FIR أسهل.

11 6. Ease of synthesis. In general, an FIR filter is algebraically more difficult to synthesize if CAD support is not available.

من غير برامج CAD، حساب الـ FIR بالإيد أصعب جبريًا (لأن الـ IIR عنده جداول وطرق analog جاهزة).

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Each point in the comparison is a direct result of FIR having no feedback and IIR having feedback.
WHY / HOW	No feedback means no poles to go unstable, errors don't recirculate, and symmetric coefficients can give exactly linear phase; feedback means a sharp cutoff from few coefficients, but with possible instability and nonlinear phase.
EXAMPLE · LINK	So the choice is a trade-off: FIR gives quality of phase and safety, IIR gives efficiency — exactly what Example 5.1 shows in numbers.

الأسئلة المتوقعة

1 What is linear phase and why is it important?

The phase is proportional to frequency, so all components are delayed by the same time and the waveform is not distorted — no phase distortion. Important in data transmission, biomedicine, audio and images.

1.6-b "From these points a broad guideline ... can be drawn" p. 8

الريبورت بيقول إيه — جملة بجملة

1 From these points a broad guideline on when to use each type can be drawn:

من النقط الست نطلع بقاعدة عامة:

2 Use an IIR filter when the only important requirements are a sharp cutoff and high throughput, because IIR filters, especially those using elliptic characteristics, give fewer coefficients than FIR filters.

IIR لو كل اللي يهمك cutoff حاد و**throughput** عالي (عينات كتير في الثانية) — لأنه أقل coefficients، والـ **elliptic** بالذات أقلهم.

3 Use an FIR filter if the number of filter coefficients is not too large and, in particular, if little or no phase distortion is desired.

FIR لو عدد الـ coefficients معقول، وخصوصًا لو **مش عايز phase distortion**.

4 Newer DSP processors also have architectures tailored to FIR filtering, and some are designed specifically for FIR filters.

والـ DSP processors الحديثة معمولة بحيث تنفّذ الـ FIR بسرعة (multiply-accumulate في cycle واحدة) ← عيب كثرة الـ coefficients بقى أقل.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The guideline turns the comparison into a decision rule.
WHY / HOW	If what matters is only a sharp cutoff and speed, IIR wins because it needs fewer coefficients; if the waveform must not be distorted, we need linear phase, which only FIR gives exactly.
EXAMPLE · LINK	In Example 5.3 the biomedical filter needs minimal distortion, so following this rule we chose FIR.

الأسئلة المتوقعة

1 Which IIR characteristic gives the fewest coefficients?

Elliptic.

Ex 5.1-a Example 5.1 (a) — type of each filter p. 8

الريبورت بيقول إيه — جملة بجملة

1 Two filters meet identical amplitude–frequency response specifications.

عندنا فلترين بيحققوا **نفس** مواصفات الـ amplitude — فنقدر نقارنهم بعدل.

2 Filter (1) has the transfer function $H(z) = (a0 + a1z^{-1} + a2z^{-2})/(1 + b1z^{-1} + b2z^{-2})$, with a0 = 0.498, a1 = 0.927, a2 = 0.498, b1 = −0.674, b2 = −0.363.

الفلتر الأول: بسط ومقام من الدرجة التانية. لاحظ $a_0 = a_2$.

3 Filter (2) has $H(z) = \Sigma_0^{11} h(k)z^{-k}$, with h(0) = h(11), h(1) = h(10), …, h(5) = h(6).

الفلتر التاني: 12 coefficient، و**متماثلين** حوالين النص (h(0)=h(11)...) ← ده بيدّي linear phase.

4 (a) Filter (1) is an IIR filter, because its denominator contains feedback coefficients. Filter (2) is an FIR filter, because it has a finite number of coefficients (12 values of h(k)).

إزاي نعرف النوع؟ **المقام** فيه b_1, b_2 = feedback ← IIR. مفيش مقام وعدد h(k) محدود ← FIR (12).

اشرحها كده عشان يبان إنك فاهم

THE IDEA We recognize the type from the transfer function itself.

WHY / HOW A denominator with b_1 and b_2 means feedback, so filter (1) is IIR; filter (2) is just a finite sum of 12 h(k) terms, so it is FIR.

EXAMPLE · LINK Also the FIR coefficients are symmetric, h(0) = h(11), h(1) = h(10) …, and that symmetry is what gives it linear phase.

الأسئلة المتوقعة

1 What is the order of filter (1)?

2 — the highest power of z^{-1} in the denominator.

Ex 5.1-b,c Example 5.1 (b), (c) — block diagrams, difference equations, Table 1 p. 9–10

الريبورت بيقول إيه — جملة بجملة

1 (b) The block diagrams are shown in Fig. 2. The difference equations of filter (1) are w(n) = x(n) − b1w(n−1) − b2w(n−2), y(n) = a0w(n) + a1w(n−1) + a2w(n−2); filter (2) is a convolution sum over its 12 coefficients.

الـ IIR بنحسبه على خطوتين بمتغير وسيط **w(n)**: الأول جزء الـ feedback (المقام) ← w(n)، وبعدين جزء البسط ← y(n).
الميزة إن w(n−1), w(n−2) بيتشاركوا في الخطوتين فبنوفّر memory. الـ FIR مجرد Σ h(k)x(n−k) على 12 حد.

2 (c) Examination of the two difference equations gives the requirements in Table 1.

بنعدّ من المعادلات: كل coefficient = ضربة، كل N حد = N−1 جمع، والتخزين = coefficients + عينات متخزنة.

3 The IIR filter is more economical in both computation and storage than the FIR filter.

النتيجة: الـ IIR أوفر في الحساب والتخزين.

4 For the same amplitude response specifications, the number of FIR coefficients (12 in this example) is typically six times the order of the IIR transfer function (2 in this case), the order being the highest power of z in the denominator.

قاعدة: عدد coefficients الـ FIR ≈ **6 × order** الـ IIR. الـ order = أعلى أس لـ z في المقام (هنا 6×2 = 12) ← 2) ✓.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Both filters meet the same amplitude specification, so counting operations shows the real cost of each type.
WHY / HOW	Every coefficient needs one multiplication and adding N products needs N−1 additions, so the FIR needs 12 and 11 while the IIR needs only 5 and 4; storage is coefficients plus stored samples, 24 against 8.
EXAMPLE · LINK	This proves the point from 1.6: the FIR needed 12 coefficients against a 2nd-order IIR — about six times the order — but in exchange it gives linear phase.

الأسئلة المتوقعة

1 Why 24 storage locations for the FIR?

12 coefficients + 12 data samples x(n) … x(n−11).

Table 1 Computational and storage requirements (Example 5.1) p. 9

Table 1 Computational and storage requirements of the filters of Example 5.1

Requirement	FIR	IIR
Number of multiplications	12	5
Number of additions	11	4
Storage locations (coefficients and data)	24	8

الرسمة / الجدول فيه إيه — جزء جزء

1. الصفوف: عدد عمليات **الضرب**، عدد عمليات **الجمع**، وعدد **أماكن التخزين** (coefficients + samples).
2. عمود **FIR**: 12 ضرب (ضرب لكل h(k))، 11 جمع (12 حاصل ضرب محتاجين 11 جمع)، 24 تخزين = 12 coefficient + 12 sample من x(n) لـ x(n−11).
3. عمود **IIR**: 5 ضرب (a_0, a_1, a_2, b_1, b_2)، 4 جمع، 8 تخزين = 5 coefficients + 3 قيم w(n), w(n−1), w(n−2).
4. الأرقام دي **لكل output sample** — يعني كل ما تيجي عينة جديدة.

اشرحها كده عشان يبان إنك فاهم

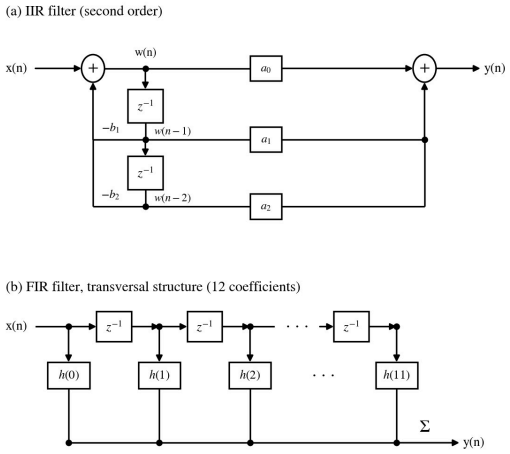
THE IDEA	Table 1 compares the cost of the two filters that meet the same amplitude specification.
WHY / HOW	The cost per output sample comes directly from the difference equations: one multiplication per coefficient and one storage location per coefficient and per stored sample, so the filter with more coefficients costs more.
EXAMPLE · LINK	The FIR needs about 2.4 times the multiplications and 3 times the storage of the IIR — this is point 4 of Section 1.6 in numbers, and the price paid for linear phase.

الأسئلة المتوقعة

1 Why are the numbers per output sample important?

In real time all of them must finish within one sampling period T = 1/Fs (Section 1.7.5).

Fig. 2 Block diagrams of the IIR and FIR filters (Example 5.1) p. 9



الرسمة / الجدول فيه إيه — جزء جزء

1. **الرموز**: الدايرة فيها + = adder، المربع z^{-1} = تأخير عينة واحدة (memory)، المربع فيه رقم = multiplier (a_0, h(k)...)، السهم = اتجاه الإشارة.
2. **(a) IIR**: الدخل x(n) بيتجمع مع خرج الفرع الراجع ($-b_1w(n-1)$ و $-b_2w(n-2)$) ← ده **w(n)** = الجزء الـ feedback. بعدين w(n), w(n−1), w(n−2) بيتضربوا في a_0, a_1, a_2 ويتجمعوا ← y(n).
3. خطوط الرجوع لشمال ($-b_1$, $-b_2$) هي اللي بتخلّيه **recursive** — فيه loop.
4. **(b) FIR**: سلسلة z^{-1} (tapped delay line) بتطلّع x(n), x(n−1)... x(n−11)، كل واحدة تتضرب في h(k) وكلهم يتجمعوا في Σ ← y(n).
5. مفيش أي خط راجع في الـ FIR ← **non-recursive**، وده سبب إنه دايمًا stable.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Fig. 2 draws the difference equations of Example 5.1 as structures, showing the computations the filter performs.
WHY / HOW	Each delay z^{-1} stores one past sample, each box is a multiplication and the adders sum the products; the IIR has feedback paths through $-b_1$ and $-b_2$, while the FIR only has forward paths.
EXAMPLE · LINK	Counting the boxes in the figure gives Table 1 directly: 5 multipliers and 2 delays for the IIR, 12 multipliers and 11 delays for the FIR.

الأسئلة المتوقعة

1 Where is the feedback in Fig. 2(a)?

The paths through $-b_1$ and $-b_2$ that bring w(n−1) and w(n−2) back to the input adder.

★ خطوات التصميم (1) — 1.7 – 1.7.2 · Part 1

1.7 Steps of Digital Filter Design — five steps p. 10

الريبورت بيقول إيه — جملة بجملة

1 The design of a digital filter involves five steps: (1) specification of the filter requirements; (2) calculation of suitable filter coefficients; (3) representation of the filter by a suitable structure (realization); (4) analysis of the effects of finite word length on filter performance; (5) implementation of the filter in software and/or hardware.

الـ 5 خطوات بالترتيب: (1) نحدد **المطلوب**. (2) نحسب **الأرقام** (coefficients). (3) نختار **شكل التوصيل** (structure) — ده معنى realization. (4) نشوف أثر **عدد الـ bits المحدود**. (5) **ننفّذ** فعلًا software أو hardware.

2 The five steps are not necessarily independent, nor always performed in the order given; to arrive at an efficient filter it may be necessary to iterate a few times between them, especially if the problem specification is not watertight.

الخطوات **مش منفصلة** ومش لازم بالترتيب ده؛ ممكن نرجع لخطوة قبلها (iterate) — مثلًا لو في خطوة 4 لقينا 16 bit مش كفاية نرجع نغيّر الـ structure. خصوصًا لو المواصفات مش «watertight» (مش محكمة / ناقصة).

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Designing a filter goes from what we want, to the numbers, to the structure, to checking the effect of finite bits, to building it.
WHY / HOW	Each step depends on the previous one, but a later step can force us back — for example, if 16-bit coefficients ruin the response, we change the structure or the coefficients.
EXAMPLE · LINK	That is why the reference says the steps may be iterated; Example 5.3 in Section 1.8 goes through all five in order.

الأسئلة المتوقعة

1 Name the five steps in order.

Specification → coefficient calculation → realization (structure) → finite word length analysis → implementation.

1.7.1-a Specification of the filter requirements — 4 items p. 10

الريبورت بيقول إيه — جملة بجملة

1 The requirement specification includes: (1) signal characteristics: the types of signal source and sink, the I/O interface, the data rates and width, and the highest frequency of interest;

خصائص الإشارة: الإشارة جاية منين (source) ورايحة فين (sink)، الـ interface (ADC/DAC...)، الـ data rate (عينات في الثانية) والـ width (عدد bits العينة)، وأعلى تردد يهمنا (بيحدد Fs).

2 (2) filter characteristics: the desired amplitude and/or phase responses and their tolerances (if any), the speed of operation and the mode of filtering (real time or batch);

خصائص الفلتر: الـ amplitude والـ phase المطلوبين وقد إيه مسموح نغلط (tolerance)، السرعة، و real-time ولا batch.

3 (3) manner of implementation: for example, as a high-level-language routine in a computer or as a DSP-processor-based system, and the choice of signal processor;

هيتنفّذ إزاي: برنامج على كمبيوتر بلغة عالية المستوى (C/MATLAB) ولا على DSP processor، وأنهي processor.

4 (4) other design constraints: for example, the cost of the filter.

قيود تانية زي التكلفة.

5 The designer may not have enough information to specify the filter completely at the outset, but as many of the requirements as possible should be specified in order to simplify the design process.

ممكن في الأول ما يبقاش عندك كل المعلومات، بس حدد أكتر حاجة ممكنة عشان التصميم يبقى أسهل وما ترجعش كتير.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The specification is the contract of the design: what goes in, what must come out, how it will be built, and at what cost.
WHY / HOW	Every later decision depends on it — the phase requirement decides FIR or IIR, the processor decides the structure and the word length.
EXAMPLE · LINK	In Example 5.3, the requirement "minimal distortion" and the TMS32010 processor in the specification are what decided the whole design.

الأسئلة المتوقعة

1 What does "mode of filtering" mean?

Whether the filter works in real time or in batch (non-real-time) mode.

1.7.1-b Tolerance scheme — Fig. 3 and the parameters δp, δs, fp, fs p. 10–11

الريبورت بيقول إيه — جملة بجملة

1 The characteristics of digital filters are often specified in the frequency domain.

المواصفات غالبًا بتتكتب بالترددات (frequency domain) — يعني «عدّي من كذا لكذا Hz وامنع من كذا».

2 For frequency-selective filters, such as lowpass and bandpass filters, the specifications are usually given as a tolerance scheme.

frequency-selective = بيختار ترددات يعدّيها وترددات يمنعها. مواصفاته بتتكتب كـ **tolerance scheme** = رسمة فيها حدود مسموحة.

3 Fig. 3 depicts such a scheme for a lowpass filter.

Fig. 3 = مثال لـ lowpass.

4 The horizontal lines indicate the tolerance limits: in the passband the magnitude response has a peak deviation of δp, and in the stopband it has a maximum deviation of δs.

الخطوط الأفقية = الحدود. في الـ **passband** الـ magnitude المفروض = 1 بس مسموح يزيد أو يقل بحد أقصى **δp**. في الـ **stopband** المفروض = 0 بس مسموح يوصل لـ **δs** كحد أقصى.

5 The width of the transition band determines how sharp the filter is; in this region the magnitude response decreases monotonically from the passband to the stopband.

transition band = المنطقة بين نهاية الـ passband وبداية الـ stopband. كل ما تبقى أضيق الفلتر يبقى **أحدّ** (sharper). فيها الـ response **بينزل باستمرار** (monotonically = من غير ما يطلع تاني).

6 The key parameters of interest are: δp: the passband deviation; δs: the stopband deviation; fp: the passband edge frequency; fs: the stopband edge frequency.

الأرقام الأربعة اللي بتحدد المواصفات: δp، δs، **fp** (آخر الـ passband)، **fs** (أول الـ stopband). ⚠ fs الصغيرة هنا مش الـ sampling frequency (اللي هي Fs الكبيرة).

اشرحها كده عشان يبان إنك فاهم

THE IDEA A real filter can't be perfect, so the tolerance scheme states how far from ideal it is allowed to be.

WHY / HOW In the passband the gain may ripple between 1 + δp and 1 − δp, in the stopband it must stay below δs, and between fp and fs the response falls; a narrower transition band means a sharper but longer filter.

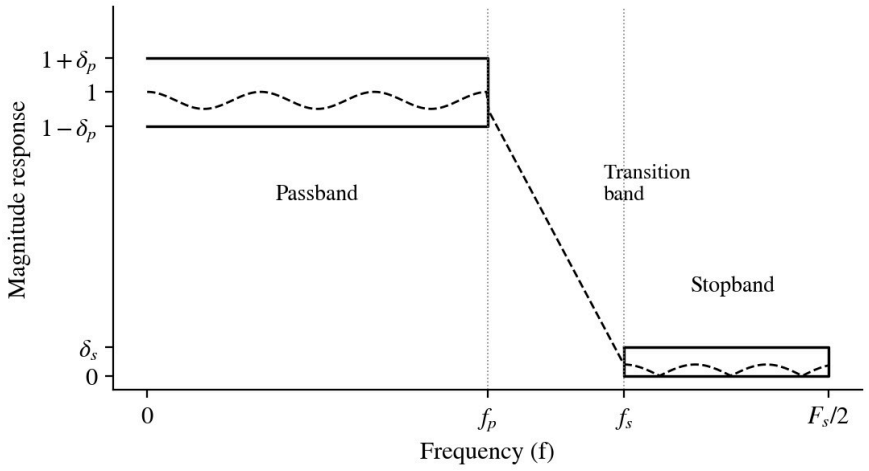
EXAMPLE · LINK Note that fs here is the stopband edge, not the sampling frequency — the axis ends at Fs/2 because that is the highest unique frequency.

الأسئلة المتوقعة

1 What are δp, δs, fp, fs?

δp: passband deviation; δs: stopband deviation; fp: passband edge frequency; fs: stopband edge frequency.

Fig. 3 Tolerance scheme for a lowpass filter p. 11



الرسمة / الجدول فيه إيه — جزء جزء

1. المحور الأفقي: التردد من 0 لـ **$F_s/2$** (أعلى تردد unique). المحور الرأسي: الـ |magnitude| الـ H|.
2. **Passband** (من 0 لـ f_p): الخطين $1+\delta_p$ و $1-\delta_p$ = الحدود المسموحة؛ الخط المتقطع المتموّج = الـ response الحقيقي (ripple) لازم يفضل بينهم.
3. **Transition band** (من f_p لـ f_s): الـ response بينزل باستمرار (monotonically)؛ مفيش شرط عليه غير إنه ينزل.
4. **Stopband** (من f_s لـ $F_s/2$): الـ response لازم يفضل **تحت δ_s**.
5. f_s الصغيرة = stopband edge، و F_s الكبيرة = sampling frequency — ماتلخبطش بينهم.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Fig. 3 states the filter specification graphically: the response must stay inside the boxes.

WHY / HOW Since an ideal brick-wall filter is impossible, we allow a ripple δp in the passband, a leakage δs in the stopband and a transition band between fp and fs; the tighter these limits and the narrower the transition band, the more coefficients we need.

EXAMPLE · LINK Example 5.2 gives these exact parameters for a bandpass filter, and Eqs. 5.5a and 5.5b convert δs and δp to decibels.

الأسئلة المتوقعة

1 Why can’t we design the ideal response (no transition band)?

An ideal brick-wall response needs an infinitely long impulse response, which can’t be implemented.

1.7.1-c Normalized frequency, decibels — Eqs. (5.5a), (5.5b); phase specification p. 11

$$A_s = -20 \log_{10} \delta_s \quad (5.5a) \quad | \quad A_p = 20 \log_{10}(1 + \delta_p) \quad (5.5b)$$

الريبورت بيقول إيه — جملة بجملة

1 The edge frequencies are often given in normalized form, that is as a fraction of the sampling frequency (f/Fs), but specifications in standard units such as hertz or kilohertz are also valid and are sometimes more meaningful, especially to an inexperienced designer.

normalized = التردد مقسوم على Fs (رقم من 0 لـ 0.5). ميزته إن التصميم ما يعتمدش على Fs. بس الـ Hz أوضح للمبتدئ.

2 The passband and stopband deviations may be expressed as ordinary numbers or in decibels, in which case they give the passband ripple and the minimum stopband attenuation respectively.

الـ δp و δs ممكن أرقام عادية أو **dB**. بالـ dB اسمهم: **passband ripple** (قد إيه بيتموّج في الـ passband) و **minimum stopband attenuation** (أقل تخميد في الـ stopband).

3 For FIR filters, the minimum stopband attenuation As and the peak passband ripple Ap, in decibels, are given by Eqs. (5.5a) and (5.5b): As = −20 log10 δs, Ap = 20 log10(1 + δp).

التحويل: $A_s = -20 \log \delta s$ (السالب عشان δs أقل من 1 فالـ log سالب، فالـ attenuation يطلع موجب). $A_p = 20 \log(1 +$ δp) (أقصى ارتفاع في الـ passband). مثال: 60 dB ← δs = 0.001.

4 The phase response of a digital filter is often not specified as meticulously as the amplitude response.

الـ phase غالبًا مش بنحدده بالتفصيل الدقيق (meticulously) زي الـ amplitude.

5 In many cases it is sufficient to indicate that phase distortion is of concern or that a linear phase response is desirable.

كفاية نقول «عايزين linear phase» أو «الـ phase distortion مهم».

6 However, when filters are used to equalize or compensate the phase response of a system, or as phase shifters, the desired phase response must be specified.

إلا لو الفلتر شغلته أصلًا الـ phase (equalizer أو phase shifter) — ساعتها لازم نحدد الـ phase بالظبط.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The same tolerances are often given in decibels, and frequencies are often normalized to the sampling frequency.
WHY / HOW	Normalizing as f/Fs makes the design independent of the actual rate, and decibels are easier to read: δs = 0.001 means the stopband is 60 dB down.
EXAMPLE · LINK	The phase is usually specified only as “linear”, unless the filter is itself a phase equalizer or phase shifter.

الأسئلة المتوقعة

1 Convert δs = 0.001 to dB.

As = $-20 \log_{10}(0.001) = 60$ dB.

Ex 5.2 Example 5.2 — specification of an FIR bandpass filter, Table 2 p. 11–12

الريبورت بيقول إيه — جملة بجملة

1 An FIR bandpass filter is to meet the following specifications: passband 0.18–0.33 (normalized), transition width 0.04 (normalized), stopband deviation 0.001 and passband deviation 0.05.

المعطى normalized: يعدّي من 0.18 لـ 0.33 من Fs، الـ transition عرضه 0.04 على كل جنب، δs = 0.001، δp = 0.05.

2 Assuming a sampling frequency of 10 kHz, the specifications in standard units are given in Table 2, and the deviations in decibels follow from Eqs. (5.5a) and (5.5b).

نضرب في Fs = 10 kHz: passband 1.8–3.3 kHz، و stopband: 0.18−0.04 = 0.14 ← 1.4 kHz الـ 0.33+0.04 = 0.37 ← stopbands هي 0–1.4 kHz و 3.7–5 kHz (5 = Fs/2). الـ dB: 60 dB و 0.42 dB.

اشرحها كده عشان يبان إنك فاهم

THE IDEA This example shows how to turn normalized specifications into real ones.

WHY / HOW We multiply every normalized frequency by Fs = 10 kHz, and the transition width 0.04 gives the stopband edges on both sides of the passband: 0.14 and 0.37, i.e. 1.4 kHz and 3.7 kHz.

EXAMPLE · LINK The deviations follow from Eq. 5.5: δs = 0.001 gives 60 dB and δp = 0.05 gives 0.42 dB.

الأسئلة المتوقعة

1 Where did 1.4 kHz and 3.7 kHz come from?

(0.18 − 0.04) × 10 kHz = 1.4 kHz and (0.33 + 0.04) × 10 kHz = 3.7 kHz.

Table 2 Specifications of Example 5.2 in standard units p. 12

Table 2 Specifications of Example 5.2 in standard units

Quantity	Value
Passband	1.8 – 3.3 kHz
Stopbands	0 – 1.4 kHz and 3.7 – 5 kHz
Stopband attenuation	$-20 \log_{10}(0.001) = 60$ dB
Passband ripple	$20 \log_{10}(1 + 0.05) = 0.42$ dB

الرسمة / الجدول فيه إيه — جزء جزء

1. **Passband**: 0.18×10 kHz = 1.8 kHz لحد 0.33×10 kHz = 3.3 kHz.
2. **Stopbands**: اتنين لأنه bandpass: (0.18−0.04)×10 = 1.4 kHz ← 0–1.4 kHz، و (0.33+0.04)×10 = 3.7 kHz ← 3.7–5 (kHz (5 = $F_s/2$).
3. **Stopband attenuation**: −20 log(0.001) = 60 dB. **Passband ripple**: 20 log(1.05) = 0.42 dB.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Table 2 converts the normalized specification of Example 5.2 into hertz and decibels.
WHY / HOW	Frequencies are multiplied by Fs = 10 kHz, the transition width is added on each side of the passband to find the stopband edges, and Eqs. 5.5a and 5.5b give the dB values.
EXAMPLE · LINK	Expressed this way the filter is easier to understand: it passes 1.8–3.3 kHz and attenuates everything else by at least 60 dB.

الأسئلة المتوقعة

1 What does 60 dB attenuation mean?

The stopband gain is at most 0.001, i.e. one thousandth of the passband.

1.7.2-a Calculation of coefficients — IIR methods: impulse invariant & bilinear p. 12

الريبورت بيقول إيه — جملة بجملة

1 In this step one of a number of approximation methods is selected and the coefficients are calculated (h(k) for an FIR filter, or ak and bk for an IIR filter) so that the filter characteristics specified in the first step are satisfied.

approximation لأن الفلتر المثالي مستحيل، فبنقرّب له بفلتر حقيقي. بنختار طريقة ونحسب الأرقام بحيث المواصفات بتاعة الخطوة 1 تتحقق.

2 The method depends on whether the filter is of the IIR or the FIR type.

الطرق مختلفة للـ IIR عن الـ FIR.

3 IIR filters. The coefficients of IIR filters are traditionally calculated by transforming known analogue filter characteristics into equivalent digital filters.

الـ IIR: الطريقة التقليدية إننا ناخد فلتر **analog معروف** ونحوّله لـ digital (عشان تصميم الـ analog قديم ومعروف وفيه جداول).

4 The two basic methods are the impulse invariant method and the bilinear transformation method.

الطريقتين الأساسيتين: **impulse invariant** و **bilinear**.

5 With the impulse invariant method, the impulse response of the original analogue filter is preserved after digitizing it, but its magnitude–frequency response is not.

impulse invariant = «الـ impulse response ما يتغيّرش»: بناخد عينات من h(t) بتاع الفلتر الـ analog ← h(n) = h(nT). فالـ time response محفوظ، بس الـ magnitude response بيتغيّر.

6 Because of inherent aliasing, the method is inappropriate for highpass or bandstop filters.

ليه؟ لأن أخذ عينات من h(t) = sampling ← لو الـ response مش محدود الترددات يحصل **aliasing** (الترددات العالية تتطوي على الواطية). الـ highpass والـ bandstop بيعدّوا ترددات عالية لحد aliasing ← ∞ شديد ← مينفعش.

7 The bilinear method, on the other hand, yields very efficient filters and is well suited to calculating the coefficients of frequency-selective filters.

bilinear بيطلّع فلاتر كفء ومناسب للـ lowpass/highpass/bandpass/bandstop.

8 It allows the design of digital filters with known classical characteristics such as Butterworth, Chebyshev and elliptic.

بيخلّينا نعمل فلاتر رقمية بخصائص كلاسيكية: **Butterworth** (ناعم مفيهوش ripple)، **Chebyshev** (أحدّ بس فيه ripple في جنب)، **elliptic** (أحدّ واحد، ripple في الجنبين).

9 Filters obtained with the bilinear method in general preserve the magnitude response characteristics of the analogue filter but not its time-domain properties.

الـ bilinear بيحافظ على **الـ magnitude** (شكل الفلتر في الترددات) بس مش على الـ time response — عكس الـ impulse invariant.

10 Efficient computer programs now exist that calculate the coefficients with the bilinear method from the filter parameters of interest.

فيه برامج بتحسب الـ coefficients بالـ bilinear على طول من المواصفات.

11 The impulse invariant method is good for simulating analogue systems, whereas the bilinear method is best for frequency-selective IIR filters.

الخلاصة: impulse invariant ← لما تكون عايز تقلّد نظام analog في الزمن. bilinear ← لفلاتر الترددات.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	For IIR filters we don't start from zero — we take a known analogue filter and convert it.
WHY / HOW	Impulse invariance samples the analogue impulse response, so the time response is kept but the spectrum overlaps (aliasing); the bilinear method maps the whole analogue frequency axis onto the digital one, so the magnitude response is kept with no aliasing.
EXAMPLE · LINK	That is why in Example 5.4 we used the bilinear method — the requirement was to preserve the amplitude response.

الأسئلة المتوقعة

1 Why is the impulse invariant method unsuitable for highpass filters?

Because of inherent aliasing: the analogue response is not band-limited, so high-frequency parts fold back (the s→z mapping $z = e^{sT}$ is many-to-one, Section 2.6).

1.7.2-b Pole-zero placement method p. 12

الريبورت بيقول إيه — جملة بجملة

1 The pole-zero placement method is an alternative approach for IIR filters.

طريقة تالتة للـ IIR: نحط الـ **poles** والـ **zeros** بإيدنا في الـ z-plane.

2 It is an easy way of calculating the coefficients of very simple filters.

سهلة للفلاتر البسيطة جدًا (زي notch يشيل تردد واحد: zero على الدايرة عند التردد ده).

3 For filters with a good amplitude response it is not recommended, because it relies on trial-and-error shuffling of the pole and zero positions.

بس مش كويسة لو عايز response دقيق، لأنها **تجربة وخطأ**: تحرّك الـ poles والـ zeros وتشوف.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Pole-zero placement designs the filter directly by putting poles and zeros in the z-plane.
WHY / HOW	A pole near the unit circle boosts the frequency at its angle, and a zero on the circle removes it, so simple filters such as a notch can be designed by hand.
EXAMPLE · LINK	For a precise amplitude response it is not recommended, because moving poles and zeros by trial and error can't meet tight tolerances.

الأسئلة المتوقعة

1 What is the idea of pole-zero placement?

A pole near the unit circle at angle Ω boosts that frequency; a zero on the unit circle removes it.

1.7.2-c FIR methods — window, frequency sampling, optimal; Table 3 p. 12–13

الريبورت بيقول إيه — جملة بجملة

1 FIR filters. There are also several methods of calculating FIR coefficients; the reference discusses three: the window method, the frequency sampling method and the optimal method (the Parks–McClellan algorithm).

الـ FIR ليه 3 طرق في الكتاب: **window**، **frequency sampling**، **optimal** (Parks–McClellan).

2 The window method offers a very simple and flexible way of computing the coefficients, but it does not give the designer adequate control over the filter parameters.

window: ناخد الـ h(n) المثالي (لا نهائي) ونقطعه بـ window (Hamming، Hanning...). سهلة، بس ما تقدرش تتحكم بالظبط في δp, δs والـ edges.

3 The main attraction of the frequency sampling method is that it allows a recursive realization of FIR filters, which can be computationally very efficient; however, it lacks flexibility.

frequency sampling: نحدد قيم الـ response عند ترددات معيّنة ونحسب منها h(n). ميزتها إنها تسمح بتنفيذ **recursive** للـ FIR (حسابات أقل)، بس مش مرنة.

4 The optimal method is now widely used in industry and, for most applications, yields the desired FIR filter.

optimal: بيحسب أحسن coefficients تحقق المواصفات بأقل عدد (الخطأ موزّع بالتساوي = equiripple). الأكتر استخدامًا في الصناعة.

5 For FIR filters it should therefore be the method of first choice, unless the particular application dictates otherwise or a CAD facility is unavailable.

فهو **الاختيار الأول** للـ FIR، إلا لو التطبيق محتاج حاجة تانية أو مفيش برنامج CAD (لأنه محتاج كمبيوتر).

6 Table 3 summarizes the six methods.

Table 3 = ملخص الطرق الست.

7 The method is chosen to suit the particular application, and the choice is influenced mainly by the critical requirements in the specification.

الطريقة بتتختار حسب التطبيق، وأهم حاجة المتطلبات **الحرجة** في المواصفات.

8 In general, the crucial choice is really between FIR and IIR.

القرار الحقيقي هو FIR ولا IIR — بعدها الطريقة تقريبًا محسومة.

9 In most cases, if the FIR properties are vital then the optimal method is a good candidate, whereas if IIR properties are desirable then the bilinear method will in most cases suffice.

FIR ← optimal. IIR ← bilinear غالبًا كفاية.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The three FIR methods trade simplicity against control over the result.
WHY / HOW	The window method is simple but does not let us control the ripple exactly, frequency sampling is efficient but inflexible, and the optimal method computes the best coefficients for given tolerances.
EXAMPLE · LINK	So the real decision is FIR or IIR first; then the optimal method for FIR and the bilinear method for IIR — as in Examples 5.3 and 5.4.

الأسئلة المتوقعة

1 What is the drawback of the window method?

It does not give the designer adequate control over the filter parameters.

Table 3 Methods of calculating filter coefficients p. 13

Table 3 Methods of calculating filter coefficients

Type	Method	Main remarks
IIR	Impulse invariant	Preserves the impulse response but not the magnitude response; inherent aliasing; inappropriate for highpass or bandstop filters; good for simulating analogue systems.
IIR	Bilinear transformation	Very efficient; well suited to frequency-selective filters; allows classical Butterworth, Chebyshev and elliptic characteristics; preserves the magnitude response but not the time-domain properties.
IIR	Pole-zero placement	Easy for very simple filters; relies on trial and error; not recommended for filters with a good amplitude response.
FIR	Window	Very simple and flexible; does not give adequate control over the filter parameters.
FIR	Frequency sampling	Allows a recursive realization that can be computationally very efficient; lacks flexibility.
FIR	Optimal (Parks–McClellan)	Widely used in industry; yields the desired FIR filter for most applications; first choice unless the application dictates otherwise or CAD is unavailable.

الرسمة / الجدول فيه إيه — جزء جزء

1. الجدول 3 أعمدة: **النوع** (IIR/FIR)، **الطريقة**، و**أهم ملاحظة** عليها.
2. الـ 3 IIR طرق): impulse invariant (بيحافظ على h(t) — فيه aliasing)، bilinear (بيحافظ على الـ magnitude — الأفضل للـ frequency-selective)، pole-zero placement (للفلاتر البسيطة بس).
3. الـ 3 FIR طرق): window (بسيطة بس تحكم قليل)، frequency sampling (كفء بس مش مرنة)، optimal Parks–McClellan (الاختيار الأول).
4. اقرأ الجدول كده: **لكل طريقة: بتحافظ على إيه، وعيبها إيه، وتنفع إمتى**.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Table 3 summarizes the six coefficient-calculation methods, three for IIR and three for FIR.

WHY / HOW Each method trades something: the IIR methods differ in what they preserve from the analogue filter (time response or magnitude), and the FIR methods differ in how much control they give over the result versus simplicity.

EXAMPLE · LINK The table supports the conclusion of 1.7.2: bilinear for IIR, optimal for FIR — which are the methods used in Examples 5.4 and 5.3.

الأسئلة المتوقعة

1 Which method would you use for a highpass IIR filter, and why not the other?

Bilinear; impulse invariance would alias because a highpass response is not band-limited.

★ خطوات التصميم (2) — Part 1 · 1.7.3 – 1.7.5

1.7.3-a Realization — IIR structures: direct, cascade, parallel (Fig. 4) p. 13–14

الريبورت بيقول إيه — جملة بجملة

1 Realization means converting a given transfer function H(z) into a suitable filter structure.

realization = نحوّل المعادلة H(z) لـ **شكل توصيل** (structure) من adders و multipliers و delays.

2 Block or flow diagrams are often used to depict filter structures, and they show the computational procedure for implementing the filter.

بنرسم الـ structure كـ block diagram بيوضّح الحسابات بالترتيب.

3 The structure used depends on whether the filter is IIR or FIR.

الـ structures مختلفة للـ IIR والـ FIR.

4 IIR structures. Three structures are commonly used: the direct form, the cascade form and the parallel form (Fig. 4).

للـ IIR تلاتة: **direct**، **cascade**، **parallel** (Fig. 4).

5 The direct form is simply a straightforward representation of the IIR transfer function.

direct = ننفّذ H(z) زي ما هي، بسط ومقام كاملين.

6 In the cascade form, the transfer function of Eq. (5.4b) is factored and expressed as the product of second-order sections.

cascade = نحلّل H(z) لعوامل (factorize) ← حاصل ضرب sections من الدرجة التانية، متوصّلين **ورا بعض**.

7 In the parallel form, H(z) is expanded using partial fractions as the sum of second-order sections.

parallel = نفكّ H(z) بالـ **partial fractions** ← مجموع sections، كلهم بياخدوا نفس الدخل والخرج = مجموعهم.

8 The parallel and cascade structures are the most widely used for IIR filters, because they lead to simpler filtering algorithms and are far less sensitive to the effects of implementing the filter with a finite number of bits than the direct structure.

الـ cascade والـ parallel الأكتر استخدامًا: الحسابات أبسط، وأقل حساسية لأخطاء الـ bits.

9 The direct structure suffers from severe coefficient sensitivity problems, especially for high-order filters, and should be avoided in these cases.

coefficient sensitivity = تغيير صغير في coefficient (من التقريب) بيغيّر مكان الـ poles كتير. في الـ direct كل الـ poles في كثيرة حدود واحدة فأي تقريب بيأثر على الكل ← ممكن pole يطلع بره الدايرة ← unstable. فنتجنّبه في الـ high order.

اشرحها كده عشان يبان إنك فاهم

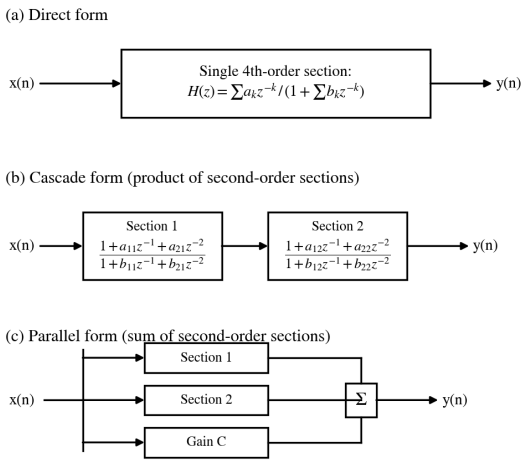
THE IDEA	The same H(z) can be built in different structures, and the choice matters because coefficients are stored with finite bits.
WHY / HOW	In the direct form one long polynomial holds all poles, so a small error in any coefficient moves all of them; in cascade or parallel form each 2nd-order section controls only its own pair of poles.
EXAMPLE · LINK	That is why Example 5.4 realizes the filter in cascade and parallel, not in direct form.

الأسئلة المتوقعة

1 **Difference between cascade and parallel?**

Cascade: H(z) factored into a product of second-order sections. Parallel: H(z) expanded by partial fractions into a sum of second-order sections.

Fig. 4 Realization structures of a fourth-order IIR filter p. 14



الرسمة / الجدول فيه إيه — جزء جزء

1. **(a) Direct**: بلوك واحد = H(z) كلها من الدرجة الرابعة: $\Sigma a_k z^{-k}/(1+\Sigma b_k z^{-k})$.
2. **(b) Cascade**: two second-order sections ورا بعض (series)؛ خرج Section 1 هو دخل = Section 2 ← H(z) = $H_1(z)\cdot H_2(z)$.
3. **(c) Parallel**: الدخل x(n) بيروح لـ Section 1 و Section 2 و Gain C في نفس الوقت، والخرج = مجموعهم في Σ ← H(z) = $H_1(z)+H_2(z)+C$.
4. كل section ليه a_{1k}, a_{2k}, b_{1k}, b_{2k} بتوعه — الـ index التاني = رقم الـ section.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Fig. 4 shows three ways to build the same fourth-order IIR filter.
WHY / HOW	The direct form uses one high-order polynomial, so a small error in any coefficient moves all four poles; cascade (product) and parallel (sum) split it into second-order sections, each controlling only two poles, so they are much less sensitive to finite word length.
EXAMPLE · LINK	Example 5.4 uses exactly forms (b) and (c): cascade by factorizing H(z), parallel by partial fractions.

الأسئلة المتوقعة

1 **How do you get the cascade form from H(z)? And the parallel?**

Cascade: factorize numerator and denominator into second-order factors. Parallel: partial-fraction expansion.

1.7.3-b FIR structures — transversal, frequency sampling, fast convolution p. 14–15

الريبورت بيقول إيه — جملة بجملة

1 FIR structures. The most widely used FIR structure is the direct form, because it is particularly simple to implement.

الـ FIR أشهر structure هو الـ **direct** لأنه بسيط جدًا.

2 It is sometimes called a tapped delay line or transversal filter, because it resembles a tapped delay line (Fig. 2(b)).

اسمه **tapped delay line**: سلسلة (z^{-1}) delays و«بناخد» (tap) من كل نقطة ونضربها في h(k). و**transversal** = نفس المعنى.

3 Two other FIR structures are also used: the frequency sampling structure and the fast convolution technique.

فيه كمان structureين: frequency sampling و fast convolution.

4 Compared with the transversal structure, the frequency sampling structure can be computationally more efficient because it leads to fewer coefficients, but it may not be as simple to implement and requires more storage.

frequency sampling: coefficients أقل ← حسابات أقل، بس أصعب في التنفيذ ومحتاج memory أكتر.

5 Fast convolution uses the computational advantage of the FFT and is particularly attractive in situations where the power spectrum of the signal is also required.

fast convolution: بدل الـ convolution في الزمن، نعمل FFT ونضرب في الترددات ونرجع بـ IFFT (أسرع للفلاتر الطويلة). وبما إننا عملنا FFT أصلًا، بنطلّع الـ **power spectrum** ببلاش.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The FIR filter is usually built as a tapped delay line: delay the input, tap every delayed sample, multiply by h(k) and add.
WHY / HOW	It is the direct picture of the convolution sum, so it is the simplest to implement; frequency sampling and fast convolution are alternatives when efficiency matters more.
EXAMPLE · LINK	Fig. 2(b) for Example 5.1 is exactly this transversal structure with 12 taps.

الأسئلة المتوقعة

1 Why is it called a tapped delay line?

The input passes through a chain of delays z^{-1}, and each delayed sample is tapped and multiplied by h(k).

1.7.3-c Lattice structure and choosing a structure p. 15

الريبورت بيقول إيه — جملة بجملة

1 Lattice structure. There are many other practical structures, but most are popular only in specific application areas.

فيه structures تانية كتير بس كل واحد مشهور في مجال معيّن.

2 An example is the lattice structure, which finds use in speech processing and linear prediction applications.

lattice (شكله زي الشبكة) بيتستخدم في الكلام و **linear prediction** (توقّع العينة الجاية من اللي فاتوا — أساس ضغط الصوت في الموبايل).

3 It can represent FIR as well as IIR filters.

يمثّل FIR و IIR.

4 The basic lattice structure is characterized by a single input and a pair of outputs; from it, N-stage FIR lattice structures and, for example, a 2-stage all-pole IIR lattice structure (one with only denominator coefficients) are derived.

الـ stage الواحدة: دخل واحد وخرجين (forward و backward). نوصّل N stage ← FIR من الدرجة N. و **all-pole** = IIR مالوش غير مقام (البسط ثابت).

5 For a given filter, the choice between structures depends largely on (i) whether it is FIR or IIR, (ii) the ease of implementation and (iii) how sensitive the structure is to the effects of finite word length.

نختار الـ structure بـ 3 حاجات: نوع الفلتر، سهولة التنفيذ، والحساسية لعدد الـ bits.

اشرحها كده عشان يبان إنك فاهم

THE IDEA The lattice is another structure, built from repeated stages, each with one coefficient K.

WHY / HOW Its stage-by-stage form suits speech and linear prediction, and it can realize both FIR and all-pole IIR filters.

EXAMPLE · LINK Example 5.5 shows how the lattice coefficients K_1, K_2 are found from the same H(z) as the transversal structure.

الأسئلة المتوقعة

1 What is an all-pole filter?

An IIR filter with only denominator coefficients (numerator = constant).

1.7.4-a Finite word length effects — four sources of degradation p. 15

الريبورت بيقول إيه — جملة بجملة

1 The approximation and realization steps assume infinite or very high precision.

خطوتين 2 و 3 بيفترضوا إن الأرقام **دقيقة جدًا** (زي ما بنحسب في الكمبيوتر).

2 In an actual implementation, however, it is often necessary to represent the filter coefficients with a limited number of bits, typically 8 to 16 bits, and the arithmetic operations of the difference equation are performed using finite-precision arithmetic.

في الحقيقة الـ hardware بيخزّن كل رقم في **8 لـ 16 bit** بس، والحسابات بدقة محدودة.

3 The effect is to degrade the performance of the filter and, in some cases, to make it unusable.

النتيجة: الأداء يقل، وأحيانًا الفلتر يبوظ خالص (unusable).

4 The designer must therefore analyze these effects and choose suitable word lengths for the filter coefficients, for the filter variables (the input and output samples) and for the arithmetic operations within the filter.

فالمصمم لازم يحلل ويختار عدد bits مناسب لـ: الـ coefficients، العينات (دخل وخرج)، ونتايج الحسابات.

5 1. Input/output signal quantization. In particular, the ADC noise due to quantizing the input signal samples is significant.

quantization الدخل والخرج: الـ ADC بيقرّب كل عينة لأقرب مستوى ← خطأ = noise، وده مهم.

6 2. Coefficient quantization. This leads to deviations in the frequency response of both FIR and IIR filters, and possibly to instability in IIR filters.

تقريب الـ coefficients: الأرقام بتتغيّر شوية ← الـ frequency response يبعد عن المطلوب، وفي الـ IIR الـ poles تتحرك وممكن تطلع بره الدايرة ← instability.

7 3. Arithmetic round-off errors. Finite-precision arithmetic yields results that need additional bits. When these are quantized to the permissible word length, often by rounding, round-off noise results, which can cause undesirable effects such as instability in an IIR filter.

round-off: ضرب رقمين 16 bit بيطلّع 32 bit؛ لما نقرّبه لـ 16 تاني يحصل خطأ = round-off noise، وفي الـ IIR ممكن يعمل instability (أو limit cycles).

8 4. Overflow. This occurs when the result of an addition exceeds the permissible word length. It leads to wrong output samples and to possible instability in IIR filters.

overflow: ناتج الجمع أكبر من أكبر رقم الـ bits تقدر تمثّله ← الرقم «يلف» ويطلع غلط (موجب كبير يبقى سالب).

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Design assumes exact numbers, but hardware stores everything with a limited number of bits — this step checks if that still works.
WHY / HOW	Errors come from four places: quantizing the input, rounding the coefficients, rounding results of multiplications, and overflow when a sum is too large; in IIR filters these errors are fed back, so they can even cause instability.
EXAMPLE · LINK	For example, rounding the b_k can push a pole that was close to the unit circle outside it, and the filter becomes unstable.

الأسئلة المتوقعة

1 Why can coefficient quantization make an IIR filter unstable?

Rounding the b_k coefficients moves the poles; a pole close to the unit circle can move outside it.

1.7.4-b Extent of degradation and when it matters p. 15–16

الريبورت بيقول إيه — جملة بجملة

1 The extent of the degradation depends on (i) the word length and the type of arithmetic used for the filtering operation, (ii) the method used to quantize the coefficients and variables to the chosen word length and (iii) the filter structure.

حجم المشكلة بيعتمد على: (1) عدد الـ bits ونوع الحساب (fixed-point ولا floating-point)، (2) طريقة التقريب (rounding ولا truncation)، (3) الـ structure.

2 From these factors the designer can assess the effects on the filter performance and take remedial action if necessary.

ومنهم المصمم يقدّر الأثر ويصلّح (يزوّد bits، يغيّر structure، يعمل scaling).

3 Depending on how the filter is implemented, some of the effects may be insignificant: when a filter is implemented as a high-level-language program on most large computers, coefficient quantization and round-off errors are not important.

على كمبيوتر كبير بـ floating point الأخطاء دي مهملة.

4 For real-time processing, finite word lengths (typically 8, 12 and 16 bits) are used to represent the input and output signals, the filter coefficients and the results of the arithmetic operations, and it is then nearly always necessary to analyze the effects of quantization on the filter performance.

بس في الـ real-time بـ 8 أو 12 أو 16 bit لازم دايمًا نحلل.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	How serious the errors are depends on the word length, the quantization method and the structure.
WHY / HOW	On a large computer with floating point the errors are tiny and can be ignored, but in real-time hardware with 8, 12 or 16 bits they must always be analyzed.
EXAMPLE · LINK	In Example 5.3 this analysis showed the 16-bit coefficients and 12-bit input were acceptable.

الأسئلة المتوقعة

1 **When can we ignore finite word length effects?**

When the filter runs as a high-level-language program on a large computer (high precision).

1.7.5-a Implementation — the basic building blocks p. 16

الريبورت بيقول إيه — جملة بجملة

1 After the coefficients have been calculated, a suitable structure chosen, and the degradation caused by quantizing the coefficients and variables verified as acceptable, the difference equation must be implemented as a software routine or in hardware.

بعد الخطوات الأربعة، ننفّذ الـ difference equation فعلًا كبرنامج أو hardware.

2 Whatever the method, the output of the filter must be computed for each sample in accordance with the difference equation (assuming a time-domain implementation).

أيًا كانت الطريقة، لكل عينة جديدة نحسب y(n) من المعادلة (لو التنفيذ في الزمن مش بالـ FFT).

3 The computation of y(n) involves only multiplications, additions/subtractions and delays, so the following basic building blocks are needed: memory (for example ROM) for storing the filter coefficients; memory (such as RAM) for storing the present and past inputs and outputs; hardware or software multiplier(s); an adder or arithmetic logic unit.

حساب y(n) = ضرب + جمع/طرح + تأخير ← محتاجين 4 حاجات: **ROM** للـ coefficients (ثابتة، بنقراها بس)، **RAM** للعينات x(n), x(n−1)... و y(n), y(n−1)... (بتتغيّر كل عينة)، **multiplier**، و**adder / ALU**.

4 The designer provides these blocks and ensures that they are suitably configured for the application.

المصمم بيوفّرهم ويظبطهم حسب التطبيق.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Implementation means actually computing y(n) for every new sample.
WHY / HOW	The difference equation needs only multiply, add and delay, so the hardware needs memory for coefficients, memory for past samples, a multiplier and an adder.
EXAMPLE · LINK	Coefficients are fixed after design so they go in ROM, while samples change every period so they go in RAM.

الأسئلة المتوقعة

1 **Why ROM for coefficients and RAM for data?**

Coefficients are fixed after design (read-only); samples change every sampling period (read/write).

1.7.5-b Batch vs real-time processing; DSP processors p. 16

الريبورت بيقول إيه — جملة بجملة

1 The configuration depends largely on whether batch (non-real-time) or real-time processing is required.

الظبط بيعتمد على: **batch** ولا **real-time**.

2 In batch processing the entire data is already available in a memory device, as when experimental data is acquired elsewhere for later analysis.

batch: الداتا كلها متسجّلة قبل كده (زي داتا تجربة) ومش مستعجلين.

3 The filter is then often implemented in a high-level language on a general-purpose computer, where all the basic blocks are already configured, so batch processing is essentially a software implementation.

فبنكتب برنامج على كمبيوتر عادي (الكمبيوتر فيه ROM و RAM و ALU جاهزين) ← software.

4 In real-time processing the filter is required either (i) to operate on the present input sample x(n) to produce the current output sample y(n) before the next input sample arrives, that is within the inter-sample period, or (ii) to operate on an input block of data, using an FFT algorithm for example, to produce an output block of data within a period proportional to the block length.

real-time بطريقتين: (1) عينة بعينة: y(n) لازم يخلص قبل ما x(n+1) توصل (في أقل من T = 1/Fs بـ block). (2) block (بالـ block: الخرج يخلص في وقت متناسب مع طول الـ block).

5 Real-time filtering may require fast and dedicated hardware if the sample rate is very high or the filter is of high order.

لو Fs عالي أو الفلتر order عالي ← محتاجين hardware سريع مخصوص.

6 For most audio-frequency work, DSP processors such as the DSP56000 (Motorola) and the TMS320C25 (Texas Instruments) are adequate and offer considerable flexibility, since they have all the basic blocks on board, including built-in hardware multiplier(s).

للصوت، الـ DSP processors (DSP56000، TMS320C25) كفاية، لأن فيهم كل البلوكات وفيهم **hardware multiplier** (بيضرب في cycle واحدة).

7 In some applications, standard 8-bit or 16-bit microprocessors such as the Motorola 6800 or 68000 families offer attractive alternatives.

وأحيانًا microprocessor عادي (6800/68000) يكفي.

8 In addition to the signal processing hardware, the designer must provide suitable input-output interfaces (for example, analogue-to-digital conversion) according to the type of data source and sink.

وغير الـ processor لازم ADC/DAC وواجهات حسب مصدر ومقصد الداتا.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The implementation depends on whether we have time: batch or real time.
WHY / HOW	In batch processing all data is already stored, so an ordinary program on a computer is enough; in real time each output must be ready before the next sample, within T = 1/Fs, so fast DSP processors with hardware multipliers are needed.
EXAMPLE · LINK	For example, at Fs = 128 Hz in Example 5.3 the processor has 1/128 s ≈ 7.8 ms to compute each output.

الأسئلة المتوقعة

1 Define real-time processing.

The output y(n) is produced from x(n) before the next input sample arrives, i.e. within the sampling period T = 1/Fs.

★ الأمثلة المحلولة — Part 1 · 1.8

Ex 5.3 Example 5.3 — the five steps applied to a biomedical filter (Table 4) p. 17

الريبورت بيقول إيه — جملة بجملة

1 A digital filter is required for real-time physiological noise reduction with the following amplitude response specifications: passband 0–10 Hz, stopband 20–64 Hz, sampling frequency 128 Hz, maximum passband deviation below 0.036 dB and stopband attenuation above 30 dB.

المطلوب lowpass real-time: يعدّي 0–10 Hz، يمنع 20–64 Hz (64 = Fs/2)، Fs = 128 Hz، ripple أقل من 0.036 dB، وattenuation أكتر من 30 dB.

2 Other important requirements are that (1) minimal distortion of the harmonic relationships between the in-band signal components is highly desirable, (2) the time available for filtering is limited because the filter is part of a larger process, and (3) the filter is implemented with the Texas Instruments TMS32010 DSP processor with the analogue input digitized to 12 bits.

(1) «minimal distortion of the harmonic relationships» = المركّبات اللي جوه الـ passband ما تتزحزحش بالنسبة لبعض ← يعني **linear phase**. (2) وقت قليل للحساب. (3) المعالج TMS32010 والـ ADC بـ 12 bit.

3 Step 1 — Requirement specification: the role and performance requirements of the filter, together with the important constraints, are those stated in the problem.

الخطوة 1 محلولة خلاص: هي المعطيات.

4 Step 2 — Coefficients: minimal distortion and limited processing time are best achieved with a linear phase FIR filter, with the coefficients obtained using the optimal method.

FIR ← linear phase. ووقت قليل ← أقل عدد coefficients ← **optimal method**.

5 Step 3 — Structure: the transversal structure leads to the most efficient implementation using the TMS32010 processor.

الـ **transversal** لأن الـ TMS32010 عنده تعليمات بتعمل «ضرب وجمع وتأخير» مع بعض فبيبقى أسرع.

6 Step 4 — Finite word length: fixed-point arithmetic with 16-bit coefficients; input quantization noise and coefficient-quantization deviation are insignificant; a 32-bit accumulator with rounding only the final sum makes round-off negligible; each coefficient is divided by Σ|h(k)| before quantizing to avoid overflow.

16 bit لكل coefficient كفاية. حواصل الضرب بتتجمع في **accumulator 32 bit** ونقرّب مرة واحدة في الآخر (بدل كل ضربة) ← round-off مهمل. ونقسم كل coefficient على Σ|h(k)| ← أقصى خرج ممكن = أقصى دخل ← مفيش overflow.

7 Step 5 — Implementation: design and configure the TMS32010-based hardware with the necessary interfaces, then write a program that handles the I/O and calculates y(n) for each new x(n).

نعمل الـ hardware + برنامج يقرا x(n) ويحسب y(n) لكل عينة.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	This example applies the five design steps to one real problem, and each decision comes from the specification.
WHY / HOW	“Minimal distortion of harmonic relationships” means no phase distortion, so a linear-phase FIR; the TMS32010 favours the transversal structure; and 16-bit coefficients with a 32-bit accumulator and scaling by Σ|h(k)| keep the finite-word-length errors and overflow negligible.
EXAMPLE · LINK	Its numbers also connect to Part 2: the stopband ends at 64 Hz because that is Fs/2 for Fs = 128 Hz, which I use again in Table 8.

الأسئلة المتوقعة

1 Why FIR and not IIR here?

"Minimal distortion of harmonic relationships" means no phase distortion → linear phase → FIR.

Table 4 The five design steps applied to Example 5.3 p. 17

Table 4 Application of the five design steps to Example 5.3

Design step	Decision and reasoning
1. Requirement specification	The role and performance requirements of the filter, together with the important constraints, are those stated in the problem.
2. Coefficients	Minimal distortion and limited processing time are best achieved with a linear phase FIR filter, with the coefficients obtained using the optimal method.
3. Structure	The transversal structure leads to the most efficient implementation using the TMS32010 processor.
4. Finite word length	Fixed-point arithmetic is used with each coefficient represented by 16 bits (after rounding). The analysis showed that the input quantization noise and the frequency response deviation due to coefficient quantization are both insignificant. Using the 32-bit accumulator to sum the coefficient-data products and rounding only the final sum reduces round-off errors to negligible levels. To avoid overflow, each coefficient is divided by the sum of |h(k)| before quantizing to 16 bits.
5. Implementation	Design and configure the TMS32010-based hardware (if it does not already exist) with the necessary input/output interfaces, then write a TMS32010 program that handles the I/O protocols and calculates y(n) for each new input x(n).

الرسمة / الجدول فيه إيه — جزء جزء

1. عمود **Design step** = الخطوات الخمسة بتاعة 1.7 بالترتيب.
2. عمود **Decision and reasoning** = القرار في كل خطوة **وسببه** من المواصفات.
3. اقرأه كده: **المطلوب** ← **القرار**: أقل distortion ← linear-phase FIR (optimal)؛ معالج transversal ← TMS32010؛ 16 bit + accumulator 32 bit + scaling ← أخطاء مهملة ومفيش overflow؛ في الآخر hardware + برنامج.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Table 4 shows how the five design steps of Section 1.7 are applied to a real biomedical filter.

WHY / HOW Each decision is justified by a requirement in the specification, which shows that the steps are linked: the specification decides the filter type, the processor decides the structure and the word length.

EXAMPLE · LINK It is the practical summary of the whole of Part 1 — types, choosing between them, and all the design steps in one example.

الأسئلة المتوقعة

1 Which requirement decided FIR?

Minimal distortion of the harmonic relationships → linear phase → FIR.

Ex 5.4-a Example 5.4 — analogue H(s) → IIR H(z) by the bilinear method p. 18

الريبورت بيقول إيه — جملة بجملة

1 An analogue filter with the transfer function $H(s) = 1/(s^3 + 2s^2 + 2s + 1)$ is to be converted into an equivalent digital filter operating at a sampling frequency of 256 Hz.

فلتر analog (Butterworth درجة تالتة — المقام ده هو بالظبط (Butterworth normalized عايزين نحوّله لـ digital = Fs 256 Hz.

2 The coefficients are to be obtained, and the filter is to be realized in cascade and in parallel structures.

المطلوب: الـ coefficients، وبعدين cascade و parallel.

3 1. Coefficients. To preserve the amplitude response of the analogue function, the bilinear method is used. Applying it to H(s) gives $H(z) = 0.1432(1 + 3z^{-1} + 3z^{-2} + z^{-3})/(1 - 0.1801z^{-1} + 0.3419z^{-2} - 0.0165z^{-3})$.

عايزين نحافظ على الـ amplitude ← **bilinear** (نعوّض s بدالة في z). البسط $(1+z^{-1})^3$ لأن الـ bilinear بيودّي التردد ∞ لـ z = −1. المقام درجة تالتة.

اشرحها كده عشان يبان إنك فاهم

THE IDEA This example converts an analogue filter H(s) into a digital IIR filter H(z) — it is the bridge between my two parts.

WHY / HOW The bilinear method was chosen because it keeps the amplitude response; the analogue poles −1 and −0.5 ± j0.866 are in the left half of the s-plane, so the digital poles come out inside the unit circle and the filter stays stable.

EXAMPLE · LINK So H(s) is the Laplace description and H(z) the z-transform description of the same filter, as explained in Section 2.6.

الأسئلة المتوقعة

1 Why the bilinear and not the impulse invariant method?

To preserve the amplitude (magnitude) response and avoid aliasing.

Ex 5.4-b Example 5.4 — cascade and parallel realizations p. 18

الريبورت بيقول إيه — جملة بجملة

1 2. Cascade realization. H(z) is factorized into second-order and first-order sections: $H(z) = 0.1432 \cdot (1 + 2z^{-1} + z^{-2})/(1 - 0.1307z^{-1} + 0.3355z^{-2}) \cdot (1 + z^{-1})/(1 - 0.0490z^{-1})$.

حلّلنا H(z) لـ section درجة تانية (فيها الـ poles المركّبة) × section درجة أولى (فيها الـ pole الحقيقي).

2 The block diagram follows the cascade form of Fig. 4(b), and the corresponding difference equations are: $w1(n) = 0.1432x(n) + 0.1307w1(n-1) - 0.3355w1(n-2)$; $y1(n) = w1(n) + 2w1(n-1) + w1(n-2)$; $w2(n) = y1(n) + 0.049w2(n-1)$; $y2(n) = w2(n) + w2(n-1)$.

كل section: w = الجزء الـ feedback (معاملات المقام **بإشارة عكسية** لأنها بتتنقل للناحية التانية)، و y = البسط. خرج الأولى y_1 هو دخل التانية.

3 3. Parallel realization. H(z) is expressed using partial fractions: $H(z) = (1.2916 - 0.08407z^{-1})/(1 - 0.131z^{-1} + 0.3355z^{-2}) + 7.5268/(1 - 0.049z^{-1}) - 8.6753$.

بالـ partial fractions: section تانية + أولى + **ثابت** (لأن درجة البسط = درجة المقام).

4 The block diagram follows the parallel form of Fig. 4(c), and the difference equations are: $w1(n) = x(n) + 0.131w1(n-1) - 0.3355w1(n-2)$; $y1(n) = 1.2916w1(n) - 0.08407w1(n-1)$; $w2(n) = x(n) + 0.049w2(n-1)$; $y2(n) = 7.5268w2(n)$; $y3(n) = -8.6753x(n)$; $y(n) = y1(n) + y2(n) + y3(n)$.

كل فرع بياخد نفس x(n) والخرج = مجموع التلاتة. للتأكد: 1.2916 + 7.5268 − 8.6753 = 0.1431 ≈ 0.1432 ← $z^{-1} = 0$ ✓.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Here the same H(z) is realized in two structures.

WHY / HOW For cascade we factor H(z) into a 2nd-order and a 1st-order section in series; for parallel we use partial fractions so the output is the sum of the sections plus a constant.

EXAMPLE · LINK As a check, both forms give the same value when $z^{-1} = 0$: 1.2916 + 7.5268 − 8.6753 ≈ 0.1432.

الأسئلة المتوقعة

1 How do you get the difference equation of a section?

Write the denominator part as w(n) = input − b_1w(n−1) − b_2w(n−2), then the numerator part as y(n) = a_0w(n) + a_1w(n−1) + ...

Ex 5.5 Example 5.5 — transversal and two-stage lattice of an FIR filter p. 18–19

الريبورت بيقول إيه — جملة بجملة

1 The transfer function of an FIR filter is $H(z) = 1 - 1.3435z^{-1} + 0.9025z^{-2}$.

FIR بـ 3 coefficients: h(0)=1، h(1)=−1.3435، h(2)=0.9025.

2 The realization block diagram is required for (1) the transversal structure and (2) a two-stage lattice structure, together with the lattice coefficients.

المطلوب الرسمتين + معاملات الـ lattice K_1 و K_2.

3 1. Transversal structure. The diagram follows directly from the transfer function, as in Fig. 2(b) with three coefficients: y(n) = x(n) + h(1)x(n−1) + h(2)x(n−2) (5.6).

زي Fig. 2(b) بس 3 taps: 2 delays و 3 ضربات.

4 2. Two-stage lattice structure. The outputs are related to the input by y2(n) = x(n) + K1(1 + K2)x(n−1) + K2x(n−2) and w2(n) = K2x(n) + K1(1 + K2)x(n−1) + x(n−2) (5.7).

خرج الـ lattice بعد stageين مكتوب بدلالة K_1 و K_2. لاحظ w_2 نفس المعاملات بس **بالعكس**.

5 Comparing Eq. (5.6) with Eq. (5.7a) and equating coefficients gives K1 = h(1)/(1 + h(2)) and K2 = h(2): K2 = 0.9025, K1 = −1.3435/1.9025 = −0.7062.

نقارن معامل $x(n-2)$: $K_2 = h(2) = 0.9025$. ومعامل $x(n-1)$: $K_1(1+K_2) = h(1) \leftarrow K_1 = -1.3435/1.9025 = -0.7062$.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	This example builds the same FIR filter as a transversal structure and as a two-stage lattice.
WHY / HOW	The transversal form reads h(k) directly from H(z); for the lattice we write its output equation and match its coefficients with h(1) and h(2).
EXAMPLE · LINK	Matching gives $K_2 = h(2) = 0.9025$ and $K_1 = h(1)/(1 + K_2) = -0.7062$.

الأسئلة المتوقعة

1 Derive K_1 and K_2.

Lattice: $y_2(n) = x(n) + K_1(1+K_2)x(n-1) + K_2x(n-2)$. Compare with x(n) + h(1)x(n−1) + h(2)x(n−2): $K_2 = h(2) = 0.9025$, $K_1 = h(1)/(1+K_2) = -1.3435/1.9025 = -0.7062$.

★ لابلاس: الوصف والتطبيقات — Part 2 · 2.1 – 2.3

2-intro Introduction of Part 2 — plan of the part p. 20

الريبورت بيقول إيه — جملة بجملة

1 This part presents the Laplace transform as it appears in the course material: the course textbook and the lecture notes of the DSP course.

الجزء ده بيقدّم الـ Laplace زي ما هو في الكتاب والمحاضرات بتوع المادة.

2 In this material the Laplace transform is the tool that describes continuous-time signals and systems, and the Z-transform is its discrete-time counterpart.

الـ Laplace = أداة الإشارات والأنظمة **المتصلة**. الـ Z = نفس الأداة بس لـ **متقطعة** (counterpart = المقابل).

3 The part therefore recalls first the continuous-time signals and systems that the Laplace transform describes, then describes the transform and its applications, then follows the passage from continuous time to discrete time through sampling, then presents the Z-transform, and finally shows the relationship between the two transforms.

خطة الجزء: إشارات متصلة ← Laplace وتطبيقاته ← الـ sampling (المعبر من المتصل للمتقطع) ← الـ Z ← العلاقة بينهم.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Part 2 follows the path of a signal: continuous time, then sampling, then discrete time.
WHY / HOW	The Laplace transform describes the continuous side and the z-transform the discrete side, so to relate them I need the sampling step in the middle.
EXAMPLE · LINK	This is the same path as Fig. 1 in Part 1, and it is why IIR filters can be designed from analogue ones.

الأسئلة المتوقعة

1 Why did you start with continuous-time signals?

Because they are what the Laplace transform describes; sampling then leads to discrete time and the z-transform.

2.1-a Analogue signals and the sinusoid — Eqs. (2.1), (2.2) p. 20

الريبورت بيقول إيه — جملة بجملة

1 Most signals of practical interest, such as speech, biological, seismic, radar and sonar signals, and communication signals such as audio and video, are analogue signals.

معظم الإشارات الحقيقية **analog**: الكلام، الإشارات الحيوية (ECG)، الزلازل (seismic)، الرادار والسونار، الصوت والفيديو.

2 An analogue or continuous-time signal is defined for every value of the continuous time variable t, and the subscript a in xa(t) denotes an analogue signal.

continuous-time = ليها قيمة عند **كل لحظة** t (مش عند لحظات متقطعة). والـ a في analog = $x_a(t)$.

3 The basic example is the continuous-time sinusoidal signal $x_a(t) = A\cos(\omega t + \theta)$, $-\infty < t < \infty$ (2.1), which is completely characterized by three parameters: the amplitude A, the frequency ω in rad/s and the phase θ in radians.

أبسط مثال: الـ sinusoid. بيتوصف بـ 3 أرقام بس: **A** (أقصى قيمة)، **ω** (سرعة الدوران rad/s)، **θ** (البداية / الإزاحة).

4 Instead of ω, the frequency f in cycles per second or hertz (Hz) is often used, where $\omega = 2\pi f$ (2.2). In terms of f the signal is written $x_a(t) = A\cos(2\pi f t + \theta)$.

بدل ω ممكن نستخدم **f** بالـ Hz (عدد الدورات في الثانية). كل دورة = 2π rad ← $\omega = 2\pi f$.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	An analogue signal exists at every instant of time, and the basic example is the sinusoid A cos(ωt + θ).
WHY / HOW	It is described by only three numbers: amplitude, frequency and phase — and ω = 2πf just converts cycles per second to radians per second.
EXAMPLE · LINK	For example, 50 Hz is ω = 100π rad/s.

الأسئلة المتوقعة

1 **What does the subscript a in $x_a(t)$ mean?**

Analogue signal.

2.1-b Properties of the continuous-time sinusoid p. 20

الريبورت بيقول إيه — جملة بجملة

1 The sinusoid has the following properties: for every fixed value of f the signal is periodic, xa(t + Tp) = xa(t), where Tp = 1/f is the fundamental period.

(1) **periodic دايمًا** لأي f: بتتكرر كل $T_p = 1/f$ ثانية (fundamental period = أصغر زمن تكرار).

2 For f = 0 the period Tp is infinite, which is consistent with the relation f = 1/Tp.

لو f = 0 الإشارة ثابتة (DC) ← الـ period = ∞، ومنطقي لأن 1/0 = ∞.

3 Continuous-time sinusoidal signals with different frequencies are themselves distinct.

(2) أي ترددين مختلفين = إشارتين **مختلفتين** فعلًا (هنشوف إن ده مش صحيح في المتقطع).

4 Increasing the frequency f increases the rate of oscillation of the signal, in the sense that more periods are included in a given time interval.

(3) كل ما f تزيد الإشارة تتذبذب أسرع (دورات أكتر في نفس الزمن).

5 Because the time variable t is continuous, the frequency can be increased without limit, so $-\infty < f < \infty$ and $-\infty < \omega < \infty$.

ولأن t متصل، f تقدر تزيد **لما لا نهاية** (في المتقطع هنلاقي حد أقصى).

اشرحها كده عشان يبان إنك فاهم

THE IDEA	These properties describe continuous-time sinusoids so that I can compare them later with discrete ones.
WHY / HOW	In continuous time every sinusoid is periodic, different frequencies are always different signals, and the frequency can go up without limit.
EXAMPLE · LINK	In Section 2.4 we see that all three change after sampling — that is where aliasing comes from.

الأسئلة المتوقعة

1 **Is a CT sinusoid always periodic?**

Yes, for every value of f; Tp = 1/f.

2.1-c Harmonically related complex exponentials and the Fourier series — Eqs. (2.3), (2.4) p. 20–21

الريبورت بيقول إيه — جملة بجملة

1 Sinusoidal signals and complex exponentials play a major role in the analysis of signals and systems.

الـ sinusoids والـ complex exponentials ($e^{j\omega t} = \cos \omega t + j \sin \omega t$) أساس تحليل الإشارات، لأن أي إشارة ممكن نكتبها كمجموع منهم.

2 In continuous time the basic harmonically related complex exponentials are $s_k(t) = e^{jk\Omega_0 t} = e^{j2\pi k f_0 t}$, $k = 0, \pm 1, \pm 2, \ldots$ (2.3).

harmonically related = ترددهم مضاعفات صحيحة لتردد أساسي f_0: f_0، $2f_0$، $3f_0$...

3 For each value of k, $s_k(t)$ is periodic with fundamental period $1/(kf_0) = T_p/k$, that is with fundamental frequency kf_0.

الـ harmonic رقم k ترددها kf_0 والـ period بتاعها T_p/k.

4 All the $s_k(t)$ have the common period T_p, the frequency f_0 may take any value, and all members of the set are distinct.

كلهم بيتكرروا كل T_p (الـ k-th بيعمل k دورة في T_p)، وكلهم مختلفين عن بعض.

5 A linear combination of these signals, $x_a(t) = \Sigma c_k e^{jk\Omega_0 t}$ (2.4), is periodic with fundamental period $T_p = 1/f_0$.

لو جمعناهم بأوزان c_k ← إشارة periodic بـ T_p.

6 Its representation is called the Fourier series expansion of $x_a(t)$; the complex constants c_k are the Fourier series coefficients and the signal $s_k(t)$ is called the k-th harmonic of $x_a(t)$.

ده هو **Fourier series**: c_k = قد إيه كل harmonic موجود في الإشارة.

اشرحها كده عشان يبان إنك فاهم

THE IDEA The complex exponentials $e^{jk\Omega_0 t}$ are the building blocks of periodic signals.

WHY / HOW They all repeat every Tp, so adding them with weights c_k gives another signal with period Tp — this sum is the Fourier series.

EXAMPLE · LINK The kth term is the kth harmonic; for example a square wave is built from its odd harmonics.

الأسئلة المتوقعة

1 What is the period of the k-th harmonic?

$1/(kf_0) = T_p/k$.

2.1-d Causal continuous-time systems p. 21

الريبورت بيقول إيه — جملة بجملة

1 A continuous-time system, like a discrete-time one, is causal if it produces an output only when there is an input, and all physical systems are causal.

causal = الخرج ما يظهرش قبل الدخل (النظام مبيشوفش المستقبل). كل نظام حقيقي causal.

2 The Laplace transform, described next, is the transform normally used for such continuous-time systems and signals.

والـ Laplace هو الأداة المعتادة للأنظمة دي — جملة ربط بـ 2.2.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	A causal system cannot react before it receives an input.
WHY / HOW	Real systems can't see the future, so all physical systems are causal, and this is why we use the one-sided Laplace and z-transforms starting from t = 0 or n = 0.
EXAMPLE · LINK	This leads to the Laplace transform as the standard tool for such systems.

الأسئلة المتوقعة

1 Define a causal system.

Its output at any time depends only on present and past inputs — it does not respond before the input is applied.

2.2-a Description of the Laplace transform — s = d + jω and the s-plane p. 21

الريبورت بيقول إيه — جملة بجملة

1 Continuous-time systems or signals are normally described using the Laplace transform.

الأنظمة المتصلة بنوصفها بالـ Laplace. (التعريف اللي مش مكتوب عندك: $X(s) = \int_0^\infty x(t)e^{-st}dt$ — بيحوّل الإشارة من دالة في t لدالة في s، والتفاضل يبقى ضرب في s.)

2 Its variable is the complex Laplace variable s = d + jω (2.5), where d is the real part and ω is the angular frequency.

s رقم مركّب: **d** (الجزء الحقيقي = تخميد/نمو، e^{dt}) و **ω** (التذبذب). وكتير بيتكتب σ بدل d.

3 The values of s are displayed in the s-plane, whose horizontal axis is d and whose vertical axis is jω (Fig. 5).

s-plane: الأفقي d والرأسي jω. كل نقطة = إشارة e^{st} بتتذبذب بـ ω وبتخمد (d سالب) أو تكبر (d موجب).

4 In terms of frequency response the jω axis is the most important axis of the s-plane, since on it d = 0 and its points represent the frequencies of the system.

محور **jω** أهم محور: عليه sinusoid ← d = 0 ← نقي ← لو عوّضنا s = jω في H(s) يطلع الـ **frequency response**.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The Laplace variable s = d + jω has two parts: d for growth or decay and ω for oscillation.
WHY / HOW	Because $e^{st} = e^{dt}e^{j\omega t}$, every point of the s-plane is a decaying or growing sinusoid; on the jω axis d = 0, so we get pure sinusoids — the frequency response.
EXAMPLE · LINK	That is why the jω axis is the most important axis, and in Section 2.6 it is the line that maps onto the unit circle.

الأسئلة المتوقعة

1 Write the definition of the Laplace transform.

$X(s) = \int_0^\infty x(t)\, e^{-st}\, dt$ (one-sided, for causal signals); s = d + jω. *(Not written in your report — memorise it.)*

2.2-b Transfer function H(s); link to classical analogue filters p. 21

الريبورت بيقول إيه — جملة بجملة

1 A continuous-time system is described in the Laplace domain by its transfer function H(s).

النظام المتصل = $H(s) = Y(s)/X(s)$.

2 The analogue low-pass filter of Example 5.4 is of this kind: its transfer function is $H(s) = 1/(s^3 + 2s^2 + 2s + 1)$, a ratio of polynomials in s.

مثال من ريبورتك: فلتر 5.4 — نسبة كثيرتين حدود في s. المقام = poles ← $(s+1)(s^2+s+1)$ عند -1 و $-0.5 \pm j0.866$ (كلهم شمال ← stable).

3 Classical analogue filter characteristics such as Butterworth, Chebyshev and elliptic are the kind of continuous-time characteristics that are carried over into digital filters (Section 1.7.2).

خصائص Butterworth و Chebyshev و elliptic أصلها analog (في s) وبتتنقل للرقمي — ربط بـ 1.7.2.

اشرحها كده عشان يبان إنك فاهم

THE IDEA A continuous-time system is described by its transfer function $H(s) = Y(s)/X(s)$.

WHY / HOW Laplace turns the differential equations into polynomials in s, so the system becomes a ratio of polynomials whose poles show its behaviour.

EXAMPLE · LINK Example 5.4 is such a system: its poles are all in the left half-plane, so it is stable.

الأسئلة المتوقعة

1 Define H(s).

H(s) = Y(s)/X(s), the Laplace transform of the output over that of the input (zero initial conditions) — also the Laplace transform of the impulse response.

2.2-c The z-transform as the corresponding tool for discrete-time systems p. 21

الريبورت بيقول إيه — جملة بجملة

1 The z-transform is the corresponding tool for discrete-time signals and systems.

الـ Z = نفس الأداة للمتقطع.

2 It is a convenient yet invaluable tool for representing discrete-time signals and designing discrete-time systems, and it plays a similar role in discrete-time systems to that which the Laplace transform plays in continuous-time systems.

أداة سهلة ومهمة جدًا لتمثيل وتصميم الأنظمة المتقطعة، ودورها في المتقطع = دور الـ Laplace في المتصل (الـ Laplace بيحوّل differential equations لجبر، والـ Z بيحوّل difference equations لجبر).

3 A discrete-time system is essentially a mathematical algorithm that takes an input sequence x(n) and produces an output sequence y(n); examples are digital filters, digital controllers and digital spectrum analysers.

النظام المتقطع = algorithm من x(n) لـ y(n): فلاتر رقمية، controllers رقمية، spectrum analysers.

اشرحها كده عشان يبان إنك فاهم

THE IDEA The z-transform is to discrete-time systems what the Laplace transform is to continuous ones.

WHY / HOW A digital system is an algorithm from x(n) to y(n), and the z-transform turns its difference equation into algebra in z, exactly as Laplace does with differential equations.

EXAMPLE · LINK Digital filters, controllers and spectrum analysers are all such systems.

الأسئلة المتوقعة

1 Give examples of discrete-time systems.

Digital filters, digital controllers and digital spectrum analysers.

2.3 Applications of the Laplace transform — Table 5 p. 22

الريبورت بيقول إيه — جملة بجملة

1 Within the course material, the Laplace transform is used in the following ways (Table 5).

استخدامات الـ Laplace في المادة — 5) Table 5 صفوف).

2 The first application is the basic one: the Laplace transform gives the s-domain description H(s) of the analogue filter or system.

الأول والأساسي: نوصف النظام الـ analog بـ H(s).

3 The second and third applications show why this description matters in digital filtering.

التاني والتالت بيوضّحوا ليه ده مهم للفلاتر الرقمية.

4 The analogue characteristic is known and well established, and the digital filter is obtained by transforming it.

لأن تصميم الـ analog معروف ومتجرّب، فبناخده ونحوّله رقمي بدل ما نبدأ من الصفر.

5 The two methods behave differently.

طريقتين التحويل بيتصرّفوا مختلف:

6 The impulse invariant method preserves the impulse response of the analogue filter but not its magnitude–frequency response, and is inappropriate for highpass or bandstop filters because of aliasing.

impulse invariant: بيحافظ على h(t)، مش على الـ magnitude، وفيه aliasing ← مينفعش highpass/bandstop.

7 The bilinear method preserves the magnitude response but not the time-domain properties, which is why it was chosen in Example 5.4 to preserve the amplitude response of the analogue function.

bilinear: بيحافظ على الـ magnitude ← عشان كده اخترناه في 5.4.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	In this course Laplace is used mainly to describe analogue systems and as the starting point for digital IIR filters.
WHY / HOW	Analogue filter design is mature, so we take a known H(s) and transform it into H(z) — this is possible only for IIR, because FIR has no analogue counterpart.
EXAMPLE · LINK	The clearest application is Example 5.4, and Section 2.6 shows the mapping that makes it work.

الأسئلة المتوقعة

1 Why design an IIR filter from an analogue one?

Analogue filter design is mature and well established (Butterworth, Chebyshev, elliptic), so we transform a known characteristic instead of starting from zero.

Table 5 Applications of the Laplace transform in the course material p. 22

Table 5 Applications of the Laplace transform in the course material

Application	How the Laplace (s-domain) description is used
Describing continuous-time signals and systems	Continuous-time systems and signals are normally described using the Laplace transform; an analogue system is characterized by its transfer function H(s).
Starting point of IIR filter design	IIR coefficients are traditionally calculated by transforming known analogue filter characteristics into equivalent digital filters, using the impulse invariant or the bilinear method.
Converting an analogue filter to a digital one	In Example 5.4 the analogue transfer function H(s) is converted by the bilinear method, at a sampling frequency of 256 Hz, into a digital transfer function H(z) that preserves its amplitude response.
Linking analogue and digital filter types	Analogue filters can be readily transformed into equivalent IIR digital filters; this is not possible with FIR filters, as they have no analogue counterpart.
Interpreting the digital system through the s-plane	Through the mapping $z = e^{sT}$, the frequencies of the s-plane are related to the unit circle of the z-plane (Section 2.6).

الرسمة / الجدول فيه إيه — جزء جزء

1. 5 صفوف = 5 تطبيقات، والعمود التاني = إزاي الـ s-domain بيتستخدم في كل واحد.
2. الترتيب منطقي: **نوصف** النظام بـ H(s) ← نستخدمه **نقطة بداية** لتصميم IIR ← **نحوّله** فعلًا (Example 5.4) ← ده بيوضّح **ليه الـ IIR بس** (الـ FIR مالوش analog counterpart) ← وأخيرًا **نفهم** النظام الرقمي من الـ s-plane بـ $z = e^{sT}$.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Table 5 lists how the Laplace transform is used in this course, from describing analogue systems to designing digital filters.

WHY / HOW The rows build on each other: H(s) describes the analogue filter, and because analogue design is well established we transform H(s) into a digital H(z) — which works only for IIR filters.

EXAMPLE · LINK The last row is the bridge to Section 2.6, where $z = e^{sT}$ relates the s-plane to the z-plane.

الأسئلة المتوقعة

1 Which row is demonstrated by an example in your report?

Converting an analogue filter to a digital one — Example 5.4.

★ الـ Sampling والـ Z-transform — Part 2 · 2.4 – 2.5

2.4-a Sampling: x(n) = $x_a(nT)$ — Eqs. (2.6), (2.7) p. 22–23

$$x(n) = x_a(nT) \quad (2.6) \quad | \quad t = nT = n/F_s, \quad F_s = 1/T \quad (2.7)$$

الريبورت بيقول إيه — جملة بجملة

1 To process an analogue signal by digital means it must first be converted into a sequence of numbers of finite precision, which is analogue-to-digital (A/D) conversion.

عشان نعالج إشارة analog بالكمبيوتر لازم تبقى **أرقام بدقة محدودة** = 3) A/D conversion خطوات: sampling، quantization، coding).

2 The first step of this conversion is sampling, which converts a continuous-time signal into a discrete-time signal by taking samples at discrete-time instants.

أول خطوة **sampling**: ناخد قيم الإشارة عند لحظات معيّنة بس.

3 In periodic (uniform) sampling, the type used most often in practice, the discrete-time signal x(n) is obtained from xa(t) by x(n) = xa(nT) (2.6), where T is the sampling period (sampling interval).

periodic / uniform = العينات على مسافات متساوية T. العينة رقم n = قيمة الإشارة عند t = nT.

4 Its reciprocal Fs = 1/T is the sampling rate or sampling frequency.

$F_s = 1/T$ = عدد العينات في الثانية.

5 Periodic sampling links the time variables t and n of the continuous-time and discrete-time signals through t = nT = n/Fs (2.7).

الربط بين الزمنين: t = nT.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Sampling takes the value of the analogue signal every T seconds: x(n) = $x_a(nT)$.
WHY / HOW	This turns continuous time t into the integer n through t = nT = n/Fs; it is the first step of A/D conversion, followed by quantization and coding.
EXAMPLE · LINK	For example, at Fs = 128 Hz the samples are taken every 7.8 ms.

الأسئلة المتوقعة

1 Define sampling period and sampling rate.

T is the interval between samples; Fs = 1/T is the number of samples per second.

2.4-b Relation between frequency variables — Eqs. (2.8), (2.9) p. 23

$$x(n) = A\cos(2\pi Fn + \theta), \quad F = f/F_s \quad (2.8) \quad | \quad \Omega = 2\pi F = \omega T \quad (2.9)$$

الريبورت بيقول إيه — جملة بجملة

1 As a consequence, a relationship exists between the frequency variables of the two kinds of signals.

ولأن الزمنين مربوطين، الترددات كمان هتبقى مربوطة.

2 A discrete-time sinusoid is written x(n) = A cos(Ωn + θ), where n is the sample number, Ω is the frequency in radians per sample and θ is the phase.

الـ sinusoid المتقطع: Ω بالـ **rad/sample** (قد إيه بنلف بين كل عينة والتانية).

3 With the frequency variable F, defined by Ω = 2πF in cycles per sample, it becomes x(n) = A cos(2πFn + θ) (2.8).

F بالـ **cycles/sample** (كسر من الدورة في كل عينة).

4 Sampling the analogue sinusoid at the rate Fs and comparing the result with this form shows that F and f are linearly related by F = f/Fs, or equivalently by Ω = ωT (2.9).

نعوّض $t = n/F_s$ في $A\cos(2\pi ft) \leftarrow A\cos(2\pi(f/F_s)n)$ ← نقارن ← **$F = f/F_s$** و **$\Omega = \omega T$**.

5 Eq. (2.8) can be used to find the frequency f in hertz only if the sampling frequency Fs is known.

يعني F لوحده مايقولكش التردد الحقيقي — لازم تعرف F_s (F = 0.25 ممكن تكون 32 Hz أو 1 kHz حسب F_s).

اشرحها كده عشان يبان إنك فاهم

THE IDEA Sampling links the analogue frequency f with the digital frequency F.

WHY / HOW If I substitute t = n/Fs in A cos(2πft), I get A cos(2π(f/Fs)n), so F = f/Fs in cycles per sample and Ω = ωT in radians per sample.

EXAMPLE · LINK For example, 1 kHz sampled at 8 kHz gives F = 1/8 and Ω = π/4 — without Fs, F alone can't tell us the frequency in hertz.

الأسئلة المتوقعة

1 Derive F = f/Fs.

$x(n) = A\cos(2\pi f \cdot nT + \theta) = A\cos(2\pi(f/Fs)n + \theta) = A\cos(2\pi Fn + \theta) \Rightarrow F = f/Fs.$

2.4-c Range of frequencies — Eq. (2.10) and Table 6 p. 23

الريبورت بيقول إيه — جملة بجملة

1 The range of the frequency variable is also different.

مدى الترددات كمان مختلف.

2 For continuous-time sinusoids $-\infty < f < \infty$, as noted in Section 2.1, whereas for discrete-time sinusoids the frequencies in the range $-1/2 \le F \le 1/2$, $-\pi \le \Omega \le \pi$ (2.10) are unique.

المتصل: أي تردد. المتقطع: الترددات **الفريدة** بس من −π لـ π (أي تردد بره ده بيتكرر — هنشوف ليه).

3 In terms of the analogue frequency this means that, with sampling at the rate Fs = 1/T, the frequency f must fall in the corresponding range $-Fs/2 \le f \le Fs/2$.

نضرب في F_s: التردد الحقيقي لازم يبقى بين $-F_s/2$ و $F_s/2$.

4 The highest frequencies that can be represented are therefore fmax = Fs/2 = 1/(2T) and ωmax = π/T.

أعلى تردد نقدر نمثّله = **$F_s/2$**.

5 Table 6 summarizes the relations among the frequency variables.

Table 6 = ملخّص الوحدات والعلاقات والمدى.

اشرحها كده عشان يبان إنك فاهم

THE IDEA In discrete time only the frequencies from −π to π are unique.

WHY / HOW Since F = f/Fs and F must stay between −1/2 and 1/2, the analogue frequency must stay between −Fs/2 and Fs/2, so Fs/2 is the highest frequency we can represent.

EXAMPLE · LINK Table 6 summarizes the frequency variables, their units and their ranges for both kinds of signals.

الأسئلة المتوقعة

1 **What is the highest frequency that can be represented at Fs?**

Fs/2 = 1/(2T), i.e. $\omega_{max} = \pi/T$.

Table 6 Relations among the frequency variables p. 23

Table 6 Relations among the frequency variables

Item	Continuous-time signals	Discrete-time signals
Frequency in radians	ω (rad/s)	Ω (rad/sample)
Frequency in cycles	f (Hz)	F (cycles/sample)
Relation	$\omega = 2\pi f$	$\Omega = 2\pi F$
Link through sampling	$F = f/F_s$	$\Omega = \omega T$
Range of frequency	$-\infty < f < \infty$	$-1/2 \le F \le 1/2, -\pi \le \Omega \le \pi$

الرسمة / الجدول فيه إيه — جزء جزء

1. عمودين: **continuous-time** و **discrete-time**.
2. الوحدات: Ω (rad/sample) ↔ ω (rad/s)، و F (cycles/sample) ↔ f (Hz).
3. العلاقة جوه كل نوع: $\omega = 2\pi f$ و $\Omega = 2\pi F$. والربط بالـ sampling: $F = f/F_s$ و $\Omega = \omega T$.
4. المدى: المتصل من −∞ لـ ∞، والمتقطع $-\frac{1}{2} \le F \le \frac{1}{2}$ و $-\pi \le \Omega \le \pi$.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Table 6 collects all the frequency variables of continuous and discrete signals and how they are linked.

WHY / HOW Sampling divides frequency by Fs, so the discrete frequency is per sample instead of per second, and because DT frequencies repeat every 2π the discrete range is limited.

EXAMPLE · LINK The relation Ω = ωT in this table is the same as the angle of z in Section 2.6 — that is where the two parts of the transform relationship meet.

الأسئلة المتوقعة

1 **Why is the unit of Ω radians per sample?**

Because n counts samples, not seconds.

2.4-d Three further differences of DT sinusoids (periodicity, aliasing, max oscillation)

p. 23–24

الريبورت بيقول إيه — جملة بجملة

1 Discrete-time sinusoids differ from continuous-time ones in three further respects. First, a discrete-time sinusoid is periodic only if its frequency F is a rational number, F = k/N, and its fundamental period is N after cancelling common factors; a small change in frequency can therefore produce a large change in the period (for example F = 31/60 gives N = 60, whereas F = 30/60 gives N = 2).

(1) الـ sinusoid المتقطع يبقى periodic **بس لو F كسر** k/N (rational). والـ period = N بعد الاختصار. 30/60 = 1/2 ← N = 2، بس 31/60 مش بيتختصر ← N = 60. تغيير بسيط في التردد غيّر الـ period جامد.

2 Second, discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical, because cos[(Ω0 + 2π)n + θ] = cos(Ω0n + θ).

(2) ترددين بينهم 2π (أو مضاعفاته) = **نفس العينات بالظبط**، لأن 2πn مضاعف صحيح لـ 2π فـ cos مش بيفرق.

3 A sinusoid with |Ω| > π (or |F| > 1/2) is therefore identical to one with a frequency inside the unique range and is called an alias of it.

فأي تردد أكبر من π بيبان كأنه تردد جوه المدى = **alias** (اسم مستعار).

4 Third, the highest rate of oscillation of a discrete-time sinusoid is reached at Ω = π (F = 1/2); for Ω0 = 0, π/8, π/4, π/2 and π the sequences have the periods ∞, 16, 8, 4 and 2, so the period decreases as the frequency increases.

(3) أسرع ذبذبة عند Ω = π: العينات بتقلب (period = 2) ...+1, −1, +1. بعد π التذبذب بيبطّأ تاني (لأنه بيبقى alias).

5 To avoid the problem of aliasing, the sampling rate is selected according to the sampling theorem, Fs > 2fmax (2.11), where fmax is the largest frequency component in the analogue signal.

عشان نمنع الـ aliasing: $\mathbf{F_s > 2f_{max}}$.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Discrete sinusoids behave differently from continuous ones in three ways.
WHY / HOW	They are periodic only when F is a ratio of integers k/N; frequencies that differ by 2π give exactly the same samples (aliases); and the fastest oscillation is at Ω = π.
EXAMPLE · LINK	For example F = 30/60 = 1/2 has period 2, but F = 31/60 has period 60 — a tiny frequency change gives a huge change in the period.

الأسئلة المتوقعة

1 Is cos(0.5n) periodic?

Ω = 0.5 ⇒ F = 0.5/2π = 1/(4π), irrational ⇒ not periodic.

2.4-e Sampling theorem and Nyquist rate — Eq. (2.11) p. 24

الريبورت بيقول إيه — جملة بجملة

1 If the highest frequency contained in xa(t) is fmax = B and the signal is sampled at a rate Fs > 2B, then xa(t) can be recovered exactly from its samples.

sampling theorem: لو أعلى تردد B وأخدنا عينات أسرع من 2B ← نقدر نرجّع الإشارة الأصلية **بالظبط**.

2 The rate 2fmax is called the Nyquist rate.

Nyquist rate = $2f_{max}$.

3 For example, the major frequency components of speech fall below 3000 Hz, so fmax = 3000 Hz for the class of speech signals, and the sampling rate must exceed 6000 Hz.

مثال: الكلام أهم ترددات تحت 3 kHz ← F_s > 6 kHz (التليفون بيستخدم 8 kHz).

اشرحها كده عشان يبان إنك فاهم

THE IDEA	To avoid aliasing we must sample faster than twice the highest frequency: Fs > 2fmax.
WHY / HOW	Then every component lies inside the unique range, no two frequencies give the same samples, and the analogue signal can be recovered exactly; 2fmax is called the Nyquist rate.
EXAMPLE · LINK	For speech with fmax = 3 kHz, Fs must be more than 6 kHz; and in Example 5.3, Fs = 128 Hz covers signals up to 64 Hz.

الأسئلة المتوقعة

1 State the sampling theorem.

If the highest frequency in $x_a(t)$ is fmax = B and it is sampled at Fs > 2B, then $x_a(t)$ can be exactly recovered from its samples.

2.5-a Z-transform definition and ROC — Eqs. (2.12)–(2.14) p. 24

$$X(z) = \Sigma_{n=-\infty}^{\infty} x(n)\, z^{-n} \quad (2.12) \quad | \quad X(z) = \Sigma_{n=0}^{\infty} x(n)\, z^{-n} \quad (2.13)$$

الريبورت بيقول إيه — جملة بجملة

1 The Z-transform of a sequence x(n) that is valid for all n is defined by $X(z) = \Sigma_{-\infty}^{\infty} x(n)z^{-n}$ (2.12), where z is a complex variable.

التعريف: كل عينة x(n) مضروبة في z^{-n} ونجمع. z رقم مركّب. (z^{-n} زي «علامة» بتقول العينة دي رقمها n.)

2 In causal systems x(n) may be nonzero only for $0 \le n < \infty$, and the definition reduces to the one-sided z-transform of Eq. (2.13).

للأنظمة الـ causal مفيش عينات قبل n = 0 ← الـ Σ تبدأ من 0 = **one-sided**.

3 The region where the z-transform converges, so that the values of X(z) are finite, is called the region of convergence (ROC).

الـ Σ لا نهائية، فممكن تبقى ∞. **ROC** = قيم z اللي الـ Σ عندها بتطلع رقم محدود.

4 For the causal infinite-duration sequence of Example 4.1(d), whose samples are 1 for $n \ge 0$, the transform is a geometric series in z^{-1} with the common ratio z^{-1}.

مثال: x(n) = 1 لكل $n \ge 0$ (unit step) ← $X(z) = 1 + z^{-1} + z^{-2} + \dots$ = متتالية هندسية أساسها z^{-1}.

5 It converges when $|z| > 1$, and X(z) can then be written in the closed form $X(z) = 1/(1 - z^{-1})$ (2.14).

المتتالية الهندسية بتتجمع لو |الأساس| < 1 ← $|z^{-1}| < 1$ ← $|z| > 1$، ومجموعها $1/(1 - z^{-1})$.

6 The ROC is the exterior of the circle of unit radius centred at the origin, bounded by the circle $|z| = 1$, the radius of the pole of X(z).

الـ ROC = **برّه** دايرة نصف قطرها 1، والدايرة دي بتعدّي على الـ pole (z = 1).

7 For z = 2 the series adds up to 2, while for z = 1/2 it diverges.

تأكيد: $1 + 2 + 4 + \dots = \infty$ ← z = ½. $1 + ½ + ¼ + \dots = 2$ ← z = 2.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The z-transform turns a sequence into a function of z: each sample x(n) is multiplied by z^{-n} and summed.
WHY / HOW	The sum is infinite, so it exists only where it converges — that region is the ROC; for the unit step it is a geometric series with ratio z^{-1}, so it converges only for $\|z\| > 1$.
EXAMPLE · LINK	Check: at z = 2 the sum is 1/(1 − 1/2) = 2, but at z = 1/2 the ratio is 2 and the sum diverges.

الأسئلة المتوقعة

1 Define the ROC.

The set of z values for which the z-transform series converges (X(z) finite).

2.5-b Poles, zeros and the three ROC conclusions p. 24–25

الريبورت بيقول إيه — جملة بجملة

1 The values of z for which X(z) becomes infinite are the poles of X(z), and the values for which X(z) = 0 are its zeros.

pole = z اللي يخلّي X(z) = ∞ (المقام = 0). **zero** = z اللي يخلّي X(z) = 0 (البسط = 0).

2 From the examples of the reference three conclusions are drawn: for causal sequences of finite duration the z-transform converges everywhere except at z = 0; for causal infinite-duration sequences it converges everywhere outside a circle bounded by the radius of the pole with the largest radius; and for stable causal systems the ROC always encloses the circle of unit radius, which is important for the systems to have a frequency response.

3 استنتاجات: (1) عدد عينات **محدود** ← Σ محدود ← بيتجمع في كل حتة إلا z = 0 (لأن $z^{-n} = 1/z^n$). عدد **لا نهائي** ← بره pole. (3) دايرة أبعد نظام **stable** ← الـ ROC فيها الـ unit circle ← وده لازم عشان الـ H(z) = frequency response على الدايرة.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Poles and zeros, and the ROC, tell us the kind of sequence and whether the system is stable.
WHY / HOW	A finite causal sequence is a finite sum, so it only fails at z = 0; an infinite causal one converges outside its largest pole; and for a stable system the ROC must include the unit circle.
EXAMPLE · LINK	The unit circle matters because the frequency response is the transform evaluated on $\|z\| = 1$.

الأسئلة المتوقعة

1 Why must the ROC of a stable system include the unit circle?

The frequency response is X(z) evaluated on $\|z\| = 1$; it exists only if the transform converges there.

2.5-c Properties — linearity, delay (2.15), convolution (2.16), differentiation p. 25

الريبورت بيقول إيه — جملة بجملة

1 Several properties of the z-transform are used in DSP.

خصائص بنستخدمها كتير:

2 Linearity means that the transform of a linear combination of sequences is the same linear combination of their transforms.

linearity: $Z\{a\cdot x + b\cdot y\} = a\cdot X(z) + b\cdot Y(z)$.

3 The delay property, $x(n-m) \leftrightarrow z^{-m}X(z)$ (2.15), says that delaying a sequence by m samples multiplies its transform by z^{-m}; it is widely used to convert the transfer function of a discrete-time system into a time-domain difference equation and vice versa.

delay: تأخير m عينة = ضرب في z^{-m}. عشان كده z^{-1} في الرسومات = delay عينة واحدة، وبيها بنروح ونرجع بين H(z) والـ difference equation.

4 The convolution property, $y(n) = \Sigma\, h(k)x(n-k) \leftrightarrow Y(z) = H(z)X(z)$ (2.16), says that the convolution sum relating the output y(n) of an LTI system to its input x(n) and impulse response h(k) becomes a multiplicative process in the z-domain.

convolution: الـ convolution الصعبة في الزمن = **ضرب** عادي في z. LTI = Linear Time-Invariant.

5 H(z) is often called the system transfer function.

H(z) = Y(z)/X(z) = transfer function.

6 A fourth property, differentiation, gives the transform of n x(n) by differentiating X(z) and is useful for inverse transforms with multiple-order poles.

differentiation: $n \cdot x(n) \leftrightarrow -z\, dX(z)/dz$، مفيدة لما يكون فيه pole متكرر.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The properties are what make the z-transform useful in DSP.
WHY / HOW	The delay property turns each delay into z^{-1}, so a difference equation becomes an algebraic equation; the convolution property turns convolution into multiplication, $Y(z) = H(z)X(z)$.
EXAMPLE · LINK	This is exactly how I get H(z) from the difference equation in Eq. 2.18, which is the same as Eq. 5.4b in Part 1.

الأسئلة المتوقعة

1 Why is the delay property important?

It converts between the transfer function H(z) and the difference equation, e.g. $z^{-1}Y(z) \leftrightarrow y(n-1)$.

2.5-d Difference equation, transfer function and pole-zero form — Eqs. (2.17)–(2.19)

p. 25–26

الريبورت بيقول إيه — جملة بجملة

1 The difference equation of a discrete-time system specifies the operations the system performs, in the time domain, on the input data to produce the output.

الـ difference equation = الوصفة اللي النظام بيعملها على الدخل عشان يطلّع الخرج.

2 For most practical cases it is written as $y(n) = \Sigma\, a_k x(n-k) - \Sigma\, b_k y(n-k)$ (2.17), and using the delay property the z-domain transfer function is obtained as $H(z) = \Sigma\, a_k z^{-k}/(1 + \Sigma\, b_k z^{-k})$ (2.18).

نفس (5.3). ناخد Z للطرفين: كل $z^{-k}Y(z) \leftarrow y(n-k) \leftarrow$ نجمّع Y على شمال و X على يمين ← H(z) = (5.4b).

3 If all the denominator coefficients b_k are zero, the system output depends only on the present and past inputs and the system is an FIR system; if at least one of them is nonzero it is an IIR system.

b_k كلها صفر ← FIR. واحد بس مش صفر ← IIR.

4 If H(z) has the poles p1, …, pN and the zeros z1, …, zN, it can be written in the factored form $H(z) = K(z - z1)\ldots(z - zN)/((z - p1)\ldots(z - pN))$ (2.19), where K is the gain factor.

نحلل البسط والمقام لعوامل ← نشوف الـ zeros والـ poles على طول. K = الـ gain.

5 Poles and zeros may be real or complex, and when complex they occur in conjugate pairs so that the coefficients ak and bk are real.

لو مركّبة لازم تيجي **أزواج مترافقة** (a ± jb) عشان لما نضرب العاملين الـ z تختفي والمعاملات تطلع أرقام حقيقية.

6 The information in the z-transform is conveniently shown in a pole-zero diagram, in which X marks a pole and o marks a zero, together with the unit circle |z| = 1, which plays an important role in the analysis and design of discrete-time systems.

الـ **pole-zero diagram**: × = pole، o = zero، ومعاهم الـ unit circle (مهمة للـ stability والـ frequency response).

اشرحها كده عشان يبان إنك فاهم

THE IDEA	Using the delay property, the difference equation becomes the transfer function H(z), a ratio of two polynomials.
WHY / HOW	If all b_k are zero there is no denominator and the system is FIR; otherwise it is IIR. Factoring H(z) shows its poles and zeros, which we draw on the pole-zero diagram with the unit circle.
EXAMPLE · LINK	This is the same FIR/IIR classification as Section 1.5, now seen in the z-domain.

الأسئلة المتوقعة

1 Why do complex poles come in conjugate pairs?

So that the coefficients a_k and b_k are real.

2.5-e Stability — BIBO, Eq. (2.20) p. 26

الريبورت بيقول إيه — جملة بجملة

1 Stability. A useful stability criterion for LTI systems is that every bounded input produces a bounded output (BIBO stability).

BIBO = Bounded Input Bounded Output: أي دخل محدود يدّي خرج محدود.

2 A system is BIBO stable if and only if $\Sigma|h(k)| < \infty$ (2.20) holds, where h(k) is the impulse response.

الشرط: مجموع ||h(k)|| محدود (الـ impulse response لازم يخمد بسرعة كفاية).

3 A finite-length impulse response satisfies it automatically, so stability considerations apply only to systems with an impulse response of infinite duration.

الـ FIR: مجموع عدد محدود من الأرقام = محدود دايمًا ← stable دايمًا. فالـ stability مشكلة الـ IIR بس.

4 For the output to be bounded all the poles must lie inside the unit circle.

الشرط بالـ poles: **كلهم جوه الدايرة** (كل pole p بيدّي p^k في h(k)؛ لو |p| > 1 بيخمد).

5 A pole outside the unit circle makes the system unstable, and a pole on the unit circle is regarded as unstable or potentially unstable, except when it coincides with a zero on the unit circle.

pole برّه ← unstable. pole على الدايرة ← potentially unstable (الـ h(k) مبيخمدش)، إلا لو zero قاعد عليه ويلغيه.

6 For an unstable system the impulse response increases indefinitely with time.

النظام الـ unstable الـ h(k) بتاعه بيكبر للأبد.

7 The behaviour of a simple system with the z-transform $10z^{-1}/(1 - \alpha z^{-1})$ in the reference illustrates this: for α = 0.5 the system is stable and the impulse response decays rapidly; for α = 0.99 it is marginally stable; for α = 1 it is potentially unstable; and for α = 1.5 it is unstable and the impulse response increases rapidly.

الـ pole عند z = α: 0.5 جوه ← بيخمد بسرعة. 0.99 جوه بس قريب أوي ← بيخمد ببطء (marginally). 1 على الدايرة ← مبيخمدش. 1.5 برّه ← بيكبر.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	A system is stable if every bounded input gives a bounded output, which happens when Σ|h(k)| is finite.
WHY / HOW	An FIR h(k) is finite, so it is always stable; for IIR, each pole p gives a term like p^k in h(k), which decays only if |p| < 1 — so all poles must be inside the unit circle.
EXAMPLE · LINK	In the reference example $10z^{-1}/(1 - \alpha z^{-1})$: α = 0.5 decays (stable), but α = 1.5 grows without limit (unstable).

الأسئلة المتوقعة

1 **State the stability condition in terms of h(k) and in terms of poles.**

Σ|h(k)| < ∞; equivalently all poles of H(z) inside the unit circle.

2.5-f Impulse response — h(k) = Z^{-1}[H(z)], Example 4.11 p. 26

الريبورت بيقول إيه — جملة بجملة

1 Impulse response. The impulse response of a discrete-time system is the inverse z-transform of its transfer function, $h(k) = Z^{-1}[H(z)]$, k = 0, 1, ….

h(k) = inverse Z لـ H(z).

2 It is needed to implement an FIR system and for the stability analysis of an IIR system, and it can be used to evaluate the frequency response.

محتاجينه: (1) الـ FIR بيتنفّذ بالـ h(k) نفسها، (2) نشوف الـ IIR stable ولا لأ، (3) نحسب منه الـ frequency response.

3 It can be obtained from the power series of H(z), whose coefficients give h(k) directly, or by applying a unit impulse to the system.

طريقتين: (1) نقسم H(z) قسمة طويلة ← معاملات z^{-k} = h(k). (2) ندخّل δ(n) في الـ difference equation ونحسب.

4 In Example 4.11, $H(z) = (1 - z^{-1})/(1 + 0.5z^{-1})$ gives h(0) = 1, h(1) = −1.5, h(2) = 0.75 and h(3) = −0.375 by either method.

y(n) = x(n) − x(n−1) − 0.5y(n−1): h(0) = 1، h(1) = −1 − 0.5 = −1.5، h(2) = 0.75، h(3) = −0.375 — بيخمد لأن الـ pole −0.5 جوه.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The impulse response is the output when the input is a single impulse, and it is the inverse z-transform of H(z).
WHY / HOW	We need it to implement FIR filters and to check IIR stability; we get it by long division of H(z) or by feeding δ(n) into the difference equation.
EXAMPLE · LINK	In Example 4.11, y(n) = x(n) − x(n−1) − 0.5y(n−1) gives 1, −1.5, 0.75, −0.375 — it decays because the pole −0.5 is inside the unit circle.

الأسئلة المتوقعة

1 **Get h(n) of Ex. 4.11 by applying an impulse.**

y(n) = x(n) − x(n−1) − 0.5y(n−1). With x = δ: h(0) = 1; h(1) = −1 − 0.5(1) = −1.5; h(2) = −0.5(−1.5) = 0.75; h(3) = −0.5(0.75) = −0.375.

★ العلاقة بين لابلاس و Z — Part 2 · 2.6 – 2.8

2.6-a $z = e^{sT}$ — Eqs. (2.21), (2.22) p. 26

الريبورت بيقول إيه — جملة بجملة

1 The relationship is obtained by letting $z = e^{sT}$, where s is the complex Laplace variable of Eq. (2.5) and T is the sampling interval, T = 1/Fs.

العلاقة: $z = e^{sT}$. ليه؟ تأخير زمن T في Laplace = e^{-sT}، وتأخير عينة في $z = e^{sT}$ ← $Z = z^{-1}$.

2 Then $z = e^{sT} = e^{(d+j\omega)T} = e^{dT}e^{j\omega T}$ (2.21).

نعوّض $s = d + j\omega$ ونفصل: جزء حقيقي e^{dT} وجزء دوران $e^{j\omega T}$.

3 Thus the magnitude and the angle of z are given by $|z| = e^{dT}$, $\angle z = \omega T = 2\pi f/Fs = 2\pi\omega/\omega s$ (2.22), where ωs (rad/s) is the sampling frequency.

$|z| = e^{dT}$ (بيعتمد على d بس) و $\angle z = \omega T$ (بيعتمد على ω بس). $\omega_s = 2\pi F_s$.

4 The angle of z, ωT, is the same quantity as the discrete-time frequency Ω = ωT of Eq. (2.9): a point of the s-plane with the angular frequency ω is mapped to the point of the z-plane whose angle is the discrete-time frequency of the sampled signal.

أهم جملة: زاوية $z = \Omega$ = التردد المتقطع من (2.9). يعني التردد ω في s بيروح لنقطة زاويتها هي التردد الرقمي.

اشرحها كده عشان يبان إنك فاهم

THE IDEA	The link between the two transforms is $z = e^{sT}$.
WHY / HOW	A delay of one sampling period is e^{-sT} in Laplace and z^{-1} in the z-transform, so $z = e^{sT}$; then $\|z\| = e^{dT}$ depends on the decay d, and the angle of z is ωT.
EXAMPLE · LINK	That angle ωT is exactly the digital frequency Ω from sampling in Eq. 2.9 — so the angle on the unit circle is the frequency of the sampled signal.

الأسئلة المتوقعة

1 Derive |z| and ∠z.

$z = e^{(d+j\omega)T} = e^{dT} \cdot e^{j\omega T} \Rightarrow |z| = e^{dT}$, $\angle z = \omega T$.

2.6-b Mapping properties — Fig. 5 p. 27

الريبورت بيقول إيه — جملة بجملة

1 As ω varies from −∞ to ∞, the s-plane is mapped onto the z-plane as shown in Fig. 5.

لما ω تمشي على كل القيم، الـ s-plane كله بيتنقل للـ z-plane — Fig. 5.

2 The mapping has the following properties: the entire jω axis of the s-plane is mapped onto the unit circle of the z-plane; the left-hand half of the s-plane is mapped to the inside of the unit circle; the right-hand half of the s-plane is mapped to the outside of the unit circle.

من $d = 0$: $|z| = e^{dT}$ (محور jω) ← $|z| = 1$ (الدايرة). $d < 0$ (شمال) ← $|z| < 1$ (جوه). $d > 0$ (يمين) ← $|z| > 1$ (بره).

3 In terms of frequency response the jω axis is the most important one. On it d = 0, and the frequency points of the s-plane are related to the points of the unit circle by $z = e^{j\omega T}$ (2.23).

على محور $z = e^{j\omega T}$: jω = نقطة على الدايرة زاويتها ωT ← ده سبب إن الـ frequency response الرقمي = H(z) على الـ unit circle.

4 Table 7 shows how specific frequencies are mapped from the s-plane to the z-plane, with the angle computed from Eq. (2.22), $\omega T = 2\pi\omega/\omega s$.

Table 7 بيحسب زاوية ترددات معيّنة.

اشرحها كده عشان يبان إنك فاهم

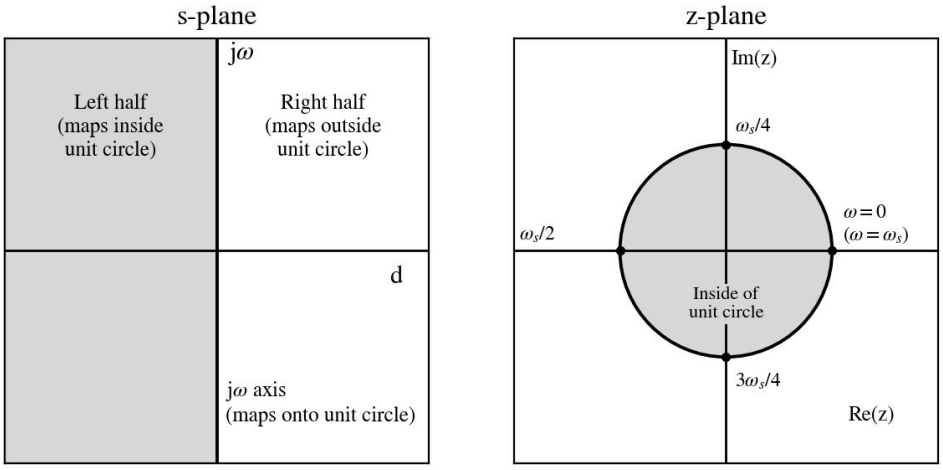
THE IDEA	From $|z| = e^{dT}$ we get the whole mapping.
WHY / HOW	On the jω axis d = 0 so |z| = 1, the unit circle; in the left half d < 0 so |z| < 1, inside; in the right half d > 0 so |z| > 1, outside.
EXAMPLE · LINK	For example ω = ωs/4 gives angle π/2, the top of the circle, and ω = ωs/2 gives angle π, the point z = −1 — as in Fig. 5.

الأسئلة المتوقعة

1 Prove the left half-plane maps inside the unit circle.

Left half ⇒ d < 0 ⇒ $|z| = e^{dT} < e^0 = 1$.

Fig. 5 Mapping of the s-plane to the z-plane p. 27



الرسمة / الجدول فيه إيه — جزء جزء

1. **الشمال: s-plane** — المحور الأفقي d، الرأسي jω. النص الشمال الرمادي (d < 0) مكتوب «maps inside unit circle»، والنص اليمين الأبيض (d > 0) «maps outside»، والمحور jω «maps onto unit circle».
2. **اليمين: z-plane** — Re(z) و Im(z) والدايرة الرمادية = الـ unit circle |z| = 1؛ جواها رمادي (نفس لون النص الشمال) عشان توضّح إنهم بيقابلوا بعض.
3. النقط على الدايرة: **ω = 0 (و ω = ωs)** عند z = 1 (يمين)، **ωs/4** فوق (z = j)، **ωs/2** شمال (z = −1)، **3ωs/4** تحت.
4. يعني لما ω تزيد من 0 لـ ωs النقطة بتلف الدايرة **مرة كاملة** عكس عقارب الساعة.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Fig. 5 shows graphically where every part of the s-plane goes under $z = e^{sT}$.

WHY / HOW Since $|z| = e^{dT}$, $d = 0$ gives $|z| = 1$, $d < 0$ gives $|z| < 1$ and $d > 0$ gives $|z| > 1$; and since the angle is $\omega T = 2\pi\omega/\omega s$, moving up the $j\omega$ axis by ωs goes once around the circle.

EXAMPLE · LINK The labelled points $\omega s/4$, $\omega s/2$, $3\omega s/4$ on the circle are the first rows of Table 7, and $\omega = 0$ and ωs sharing one point is the aliasing shown in Table 7.

الأسئلة المتوقعة

1 Why is the left half shaded the same as the inside of the circle?

To show that the left half-plane maps to the inside of the unit circle.

2.6-c Table 7 — the mapping is not unique (many-to-one) p. 27–28

الريبورت بيقول إيه — جملة بجملة

1 For example, the two frequencies $\omega = \omega s$ and $\omega = 2\omega s$ in the s-plane map to the same point on the unit circle, and $1.25\omega s$ falls on the same point as $\omega s/4$.

مثلًا $\omega_s \leftarrow 2\pi$ و $2\omega_s \leftarrow 4\pi$ نفس النقطة $z = 1$. و $2\pi + \pi/2 = 2.5\pi \leftarrow 1.25\omega_s$ نفس $\omega_s/4$.

2 This agrees with the sampling results of Section 2.4: two analogue frequencies that differ by $\omega s = 2\pi/T$ give discrete-time frequencies $\Omega = \omega T$ that differ by 2π, and discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical.

نفس اللي اتقال في 2.4: فرق ω_s في الـ analog = فرق 2π في Ω = إشارتين متطابقين.

3 Different analogue frequencies can therefore correspond to the same point of the z-plane, which is consistent with the remark in Section 1.7.2 that aliasing is inherent in the impulse invariant method.

ترددات analog مختلفة على نفس النقطة = **aliasing** — وده بالظبط سبب الـ aliasing في الـ impulse invariant (اللي بيستخدم $z = e^{sT}$).

4 The unique discrete-time frequencies, $-\pi \le \Omega \le \pi$, cover the unit circle once.

الترددات الفريدة ($-\pi$ لـ π) بتلف الدايرة **مرة واحدة**.

اشرحها كده عشان يبان إنك فاهم

THE IDEA The mapping is many-to-one: many s-plane frequencies land on the same point of the unit circle.

WHY / HOW The angle ωT repeats every 2π, so frequencies that differ by ωs give the same z — this is aliasing seen in the z-plane.

EXAMPLE · LINK Table 7 shows ωs and $2\omega s$ both at angle 2π ($z = 1$), and $1.25\omega s$ at the same point as $\omega s/4$; this is why the impulse invariant method suffers from aliasing.

الأسئلة المتوقعة

1 Why is the mapping many-to-one?

Because $e^{j\omega T}$ is periodic in ω with period $\omega s = 2\pi/T$ — each horizontal strip of width ωs in the s-plane covers the whole z-plane.

Table 7 Mapping of frequencies from the s-plane to the z-plane p. 27

Table 7 Mapping of frequencies from the s-plane to the z-plane

s-plane: ω (rad/s)	z-plane: ωT (rad)
0	0
$\omega_s/4$	$\pi/2$
$\omega_s/2$	π
$3\omega_s/4$	1.5π
ω_s	2π
$1.25\omega_s$	$\pi/2$ (+2π)
$1.5\omega_s$	π (+2π)
$1.75\omega_s$	1.5π (+2π)
$2\omega_s$	2π (+2π)

الرسمة / الجدول فيه إيه — جزء جزء

1. العمود الأول: التردد ω في الـ s-plane كجزء من ω_s. التاني: الزاوية $\omega T = 2\pi\omega/\omega_s$ على الـ unit circle.
2. من 0 لـ ω_s: الزاوية من 0 لـ 2π = لفة كاملة.
3. من $1.25\omega_s$ لـ $2\omega_s$: الزاوية بتزيد 2π تانية **(+2π)** ← نفس النقط تاني.
4. مثلًا $2\pi + \pi/2$ ← $1.25\omega_s$ = نفس نقطة $\omega_s/4$.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Table 7 computes the angle on the unit circle for different analogue frequencies.

WHY / HOW The angle is 2πω/ωs, so every increase of ωs in frequency adds 2π to the angle and returns to the same point — the mapping is many-to-one.

EXAMPLE · LINK That is aliasing: frequencies differing by ωs cannot be told apart after sampling, which is also why the impulse invariant method aliases.

الأسئلة المتوقعة

1 Compute the angle for ω = 1.5ωs.

$2\pi \times 1.5 = 3\pi \equiv \pi$, the point $z = -1$, same as ωs/2.

2.6-d Link with stability and the ROC p. 28

الريبورت بيقول إيه — جملة بجملة

1 The relationship also connects with stability.

العلاقة دي كمان بتربط الـ stability.

2 For a discrete-time system to be stable, all the poles of its transfer function must lie inside the unit circle (Section 2.5), and the left-hand half of the s-plane is mapped to the inside of the unit circle.

الرقمي poles ← stable جوه الدايرة. والنص الشمال ← جوه الدايرة.

3 Combining the two statements shows that the left-hand half of the s-plane corresponds to the region in which the poles of a stable discrete-time system lie.

النتيجة: النص الشمال (مكان poles الـ analog الـ stable) = جوه الدايرة (مكان poles الرقمي الـ stable). فالفلتر الـ analog الـ stable يتحوّل لرقمي stable.

4 In the same way the ROC of a stable causal system encloses the unit circle, which is the image of the jω axis, so that the system has a frequency response.

والـ ROC بتحتوي الدايرة (= صورة محور jω) ← النظام ليه frequency response.

اشرحها كده عشان يبان إنك فاهم

THE IDEA The mapping also connects stability in both domains.

WHY / HOW A stable analogue system has its poles in the left half-plane, the left half maps inside the unit circle, and inside the unit circle is exactly where a stable digital system needs its poles.

EXAMPLE · LINK Example 5.4 confirms it: analogue poles at −1 and −0.5 ± j0.866 gave digital poles of magnitude 0.049 and about 0.58, all inside the circle.

الأسئلة المتوقعة

1 A stable analogue filter is converted by $z = e^{sT}$. Is the digital filter stable?

Yes — its poles in the left half-plane map inside the unit circle ($|z| = e^{dT} < 1$).

2.7 Numerical illustration — Table 8 (Fs = 128 Hz) p. 28

الريبورت بيقول إيه — جملة بجملة

1 The mapping can be illustrated with the sampling frequency of Example 5.3, Fs = 128 Hz.

مثال بالأرقام بـ $F_s = 128$ Hz (من 5.3 — ربط بين الجزئين).

2 A frequency f corresponds to F = f/Fs and to the point on the unit circle at the angle ωT = Ω = 2πF, from Eqs. (2.8), (2.9) and (2.22) (Table 8).

كل تردد ← $\Omega = 2\pi F$ ← $F = f/128$ ← الزاوية على الدايرة = f.

3 The table confirms the properties found above. The frequencies between 0 and Fs/2 occupy the upper half of the unit circle, the highest unique frequency Fs/2 sits at z = −1, and a frequency larger than Fs/2 lands on a point already used by a lower frequency.

0 لـ 64 Hz ← النص اللي فوق من الدايرة. 64 Hz ($F_s/2$) ← $z = -1$. أي حاجة أكبر ← نقطة مستخدمة (160 Hz = 32 Hz).

اشرحها كده عشان يبان إنك فاهم

THE IDEA Table 8 puts real numbers on the mapping, using Fs = 128 Hz from Example 5.3.

WHY / HOW Each frequency goes to the angle Ω = 2πf/Fs, so 0 to Fs/2 fills the upper half of the circle, Fs/2 = 64 Hz reaches z = −1, and 128 Hz comes back to z = 1.

EXAMPLE · LINK 160 Hz gives 2.5π, the same point as π/2 for 32 Hz — a numerical example of aliasing.

الأسئلة المتوقعة

1 Compute the angle for 20 Hz.

F = 20/128 = 0.156; Ω = 2π × 0.156 = 0.98 rad ≈ 0.31π.

Table 8 Angle on the unit circle for Fs = 128 Hz p. 28

Table 8 Angle on the unit circle for F_s = 128 Hz

f (Hz)	$F = f/F_s$	$\Omega = 2\pi F$ (rad)	Remark
0	0	0	Point z = 1
10	0.078	0.49 (≈ 0.16π)	Passband edge of Example 5.3
20	0.156	0.98 (≈ 0.31π)	Stopband edge of Example 5.3
32	0.25	π/2	$F_s/4$
64	0.5	π	Highest rate of oscillation ($F_s/2$)
128	1	2π	Same point as 0 Hz
160	1.25	2.5π ≡ π/2	Alias of 32 Hz

الرسمة / الجدول فيه إيه — جزء جزء

1. 4 أعمدة: f بالـ Hz ← $F = f/F_s$ ← $\Omega = 2\pi F$ ← ملاحظة.
2. 10 Hz و 20 Hz = حدود الـ passband والـ stopband بتوع Example 5.3 ← زواياهم 0.16π و 0.31π.
3. π/2 ← $F_s/4$ = 32 Hz. و π (z = −1) ← $F_s/2$ = 64 Hz.
4. 2π ← 128 Hz = نفس 0 Hz. و alias = π/2 ≡ 2.5π ← 160 Hz لـ 32 Hz.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Table 8 applies the mapping to real frequencies, using Fs = 128 Hz from Example 5.3.

WHY / HOW Each row follows the chain f → F = f/Fs → Ω = 2πF, which is the angle on the unit circle; frequencies above Fs/2 come back to points already used.

EXAMPLE · LINK It ties both parts together: the passband and stopband edges of the Part 1 filter appear as points on the unit circle of Part 2.

الأسئلة المتوقعة

1 Compute the row for 10 Hz.

F = 10/128 = 0.078, Ω = 2π × 0.078 = 0.49 rad ≈ 0.16π.

2.8 Summary of the relationship — Table 9; References p. 29

الريبورت بيقول إيه — جملة بجملة

1 Table 9 compares the Laplace transform and the z-transform.

Table 9 = مقارنة سطر بسطر.

2 In summary, the Laplace transform describes continuous-time systems and signals in the s-plane, the z-transform plays the same role for discrete-time systems in the z-plane, and the substitution $z = e^{sT}$ links the two: the jω axis becomes the unit circle, the left-hand half-plane becomes its inside and the right-hand half-plane its outside.

الخلاصة: s-plane ↔ Laplace، z-plane ↔ Z، والرابط ($z = e^{sT}$ ← الدايرة، الشمال ← جوه، اليمين ← بره).

3 Through sampling, the angle of z is the discrete-time frequency Ω = ωT.

زاوية z = التردد الرقمي.

4 This link is what allows known analogue filter characteristics to be transformed into equivalent IIR digital filters.

ودة اللي بيخلّينا نحوّل فلاتر analog لـ IIR — رجوع لـ Part 1 (Example 5.4).

5 References: [1] Samir Kamal Younis, Digital Signal Processing (ELC 422), Modern Academy; [2] Samir Kamal and Sara Hassan, DSP lecture notes: Lectures 3 and 4 (Chapter 1), Lectures 9, 10-2 and 11 (Chapter 4).

المراجع: كتاب د. سمير، والمحاضرات (Chapter 1 للـ sampling، و Chapter 4 للـ z-transform).

اشرحها كده عشان يبان إنك فاهم

THE IDEA In one sentence: Laplace describes continuous systems in the s-plane, the z-transform describes discrete systems in the z-plane, and $z = e^{sT}$ connects them.

WHY / HOW Under this mapping the jω axis becomes the unit circle, the left half goes inside and the right half outside, and the angle of z is the digital frequency $\Omega = \omega T$.

EXAMPLE · LINK This connection is the reason we can convert a known analogue filter into an IIR digital filter, as in Example 5.4 — which ties Part 2 back to Part 1.

الأسئلة المتوقعة

1 Summarize the whole relationship in one sentence.

$z = e^{sT}$: the jω axis maps onto the unit circle with angle $\Omega = \omega T$, the left half-plane inside and the right half-plane outside.

Table 9 Laplace transform versus z-transform p. 29

Table 9 Laplace transform versus z-transform

Item	Laplace transform	Z-transform
Systems and signals described	Continuous-time	Discrete-time
Complex variable	$s = d + j\omega$	$z = e^{sT}$
Transfer function	H(s), e.g. the analogue filter of Example 5.4	H(z), e.g. Eqs. (5.4a) and (5.4b)
Plane and frequency axis	s-plane; the jω axis	z-plane; the unit circle
Left-hand side	Left-hand half of the s-plane	Maps to the inside of the unit circle
Right-hand side	Right-hand half of the s-plane	Maps to the outside of the unit circle
Use in filter design	Analogue filter characteristics	Digital IIR and FIR filters

الرسمة / الجدول فيه إيه — جزء جزء

1. مقارنة سطر بسطر بين الاتنين: نوع الإشارات، المتغير ($s = d + j\omega \leftrightarrow z = e^{sT}$)، الـ transfer function (H(s) زي Example 5.4 ↔ H(z) (5.4a/b زي)، محور التردد (jω ↔ unit circle)، الشمال ↔ جوه الدايرة، اليمين ↔ بره، والاستخدام في التصميم.
2. كل سطر في الجدول متشرح قبل كده في Part 2 — الجدول ده ملخّص.

اشرحها كده عشان يبان إنك فاهم

THE IDEA Table 9 summarizes Part 2 by putting the Laplace transform and the z-transform side by side.

WHY / HOW Every row is a consequence of $z = e^{sT}$: the variables, the planes, the frequency axes and the stability regions all correspond to each other.

EXAMPLE · LINK The last row links back to Part 1: analogue filter characteristics in the s-domain become digital IIR filters in the z-domain.

الأسئلة المتوقعة

1 **Which row expresses the stability link?**

Left-hand half of the s-plane ↔ inside of the unit circle.

★ كل المعادلات — Every equation

Eqs. Every numbered equation — what each symbol means

لو شاور على أي معادلة وقالك «إيه ده؟» — قول المعادلة بتعمل إيه، وكل رمز فيها معناه إيه، وجت منين. الجدول ده فيه كل معادلة مترقمة في الريبورت بالترتيب.

Eq.	Equation	Meaning of the symbols / what it says
(5.1)	$y(n) = \Sigma_{k=0}^{\infty} h(k)\, x(n-k)$	IIR as a convolution: y(n) output, x(n−k) input delayed k samples, h(k) impulse response — infinite length.
(5.2)	$y(n) = \Sigma_{k=0}^{N-1} h(k)\, x(n-k)$	FIR: same convolution but h(k) has only N values ⇒ the FIR difference equation.
(5.3)	$y(n) = \Sigma\, a_k\, x(n-k) - \Sigma\, b_k\, y(n-k)$	IIR recursive form: a_k feed-forward, b_k feedback coefficients; the y(n−k) terms are the feedback.
(5.4a)	$H(z) = \Sigma\, h(k)\, z^{-k}$	FIR transfer function: numerator only, all poles at z = 0.
(5.4b)	$H(z) = \Sigma\, a_k z^{-k} / (1 + \Sigma\, b_k z^{-k})$	IIR transfer function: zeros from the numerator, poles from the denominator.
(5.5a)	$A_s = -20 \log_{10} \delta_s$	Minimum stopband attenuation in dB from the stopband deviation (δs = 0.001 → 60 dB).
(5.5b)	$A_p = 20 \log_{10}(1 + \delta_p)$	Peak passband ripple in dB from the passband deviation (δp = 0.05 → 0.42 dB).
(5.6)	$y(n) = x(n) + h(1)x(n-1) + h(2)x(n-2)$	Transversal output of the 3-coefficient FIR of Example 5.5 (h(0) = 1).
(5.7)	$y_2(n) = x(n) + K_1(1+K_2)x(n-1) + K_2x(n-2)$	Two-stage lattice output; comparing with 5.6 gives K_2 = h(2), K_1 = h(1)/(1+h(2)). $w_2(n)$ is the second (backward) output.
(2.1)	$x_a(t) = A\cos(\omega t + \theta)$	CT sinusoid: A amplitude, ω rad/s, θ phase in radians.
(2.2)	$\omega = 2\pi f$	Converts Hz to rad/s.
(2.3)	$s_k(t) = e^{jk\Omega_0 t}$	Harmonically related complex exponentials; k-th harmonic at kf_0.
(2.4)	$x_a(t) = \Sigma\, c_k\, e^{jk\Omega_0 t}$	Fourier series: c_k coefficients, period Tp = $1/f_0$.
(2.5)	$s = d + j\omega$	Laplace variable: d real part (decay/growth), ω angular frequency.
(2.6)	$x(n) = x_a(nT)$	Periodic sampling every T seconds.
(2.7)	$t = nT = n/F_s$, $F_s = 1/T$	Links continuous time t with sample index n.
(2.8)	$x(n) = A\cos(2\pi Fn + \theta)$, $F = f/F_s$	DT sinusoid; F in cycles/sample.
(2.9)	$\Omega = 2\pi F = \omega T$	DT frequency in rad/sample — later the angle of z.
(2.10)	$-½ \le F \le ½$, $-F_s/2 \le f \le F_s/2$	Unique range of DT frequencies, so the highest representable frequency is Fs/2.
(2.11)	$F_s > 2f_{max}$	Sampling theorem; 2fmax = Nyquist rate.
(2.12)	$X(z) = \Sigma_{-\infty}^{\infty} x(n)\, z^{-n}$	Two-sided z-transform.
(2.13)	$X(z) = \Sigma_{0}^{\infty} x(n)\, z^{-n}$	One-sided z-transform for causal sequences.

(2.14)	$1 + z^{-1} + \ldots = 1/(1 - z^{-1})$, $\lvert z\rvert > 1$	Unit step: geometric series, ROC outside the pole at z = 1.
(2.15)	$x(n-m) \leftrightarrow z^{-m}X(z)$	Delay property: a delay of m samples = multiply by z^{-m}.
(2.16)	$y(n) = \Sigma\, h(k)x(n-k) \leftrightarrow Y(z) = H(z)X(z)$	Convolution property.
(2.17)	$y(n) = \Sigma\, a_k x(n-k) - \Sigma\, b_k y(n-k)$	General difference equation (same as 5.3).
(2.18)	$H(z) = Y(z)/X(z) = \Sigma\, a_k z^{-k}/(1 + \Sigma\, b_k z^{-k})$	Transfer function via the delay property (same as 5.4b).
(2.19)	$H(z) = K(z-z_1)\ldots(z-z_N)/((z-p_1)\ldots(z-p_N))$	Pole-zero form: z_i zeros, p_i poles, K gain.
(2.20)	$\Sigma_{k=0}^{\infty} \lvert h(k)\rvert < \infty$	BIBO stability condition.
(2.21)	$z = e^{sT} = e^{dT}e^{j\omega T}$	The link between the s-plane and the z-plane.
(2.22)	$\lvert z\rvert = e^{dT}$, $\angle z = \omega T = 2\pi f/F_s = 2\pi\omega/\omega_s$	Magnitude depends on d (inside/outside), angle is the DT frequency.
(2.23)	$z = e^{j\omega T}$	On the jω axis (d = 0): points on the unit circle — where the frequency response is evaluated.

★ أسئلة فخ من بره الكلام المكتوب — Trap questions

T1 Things he may ask that are NOT written in your report

1 Write the definition of the Laplace transform.

$X(s) = \int_0^{\infty} x(t)e^{-st}\,dt$, $s = d + j\omega$ (two-sided: from $-\infty$). احفظه — $s = d + j\omega$ ريبورتك فيه مفيهوش التكامل، بس.

2 What is the difference between the Laplace and Fourier transforms?

Fourier uses only $s = j\omega$ (the $j\omega$ axis, $d = 0$); Laplace uses the whole s-plane, so it handles signals/systems that grow or decay (e^{dt}).

3 The bilinear method does not use $z = e^{sT}$. What does it use?

$s = (2/T)(z - 1)/(z + 1)$. It maps the entire $j\omega$ axis onto the unit circle exactly once (one-to-one), so there is no aliasing, but the frequency axis is compressed (frequency warping), so the edge frequencies are pre-warped.

4 Laplace of $e^{-at}u(t)$ and the Z of $a^n u(n)$?

$1/(s + a)$, pole at $s = -a$; $1/(1 - az^{-1})$, pole at $z = a$. Sampling e^{-at} at $t = nT$ gives $a = e^{-aT}$ — exactly $z = e^{sT}$ applied to the pole.

5 Why are the equations in Part 1 numbered 5.x?

They keep the numbering of Chapter 5 of the course reference so they can be checked against the book; Part 2 uses its own numbering (2.x).

6 Where did you take this from? Is it your own work?

From the course book [1] and the lecture notes [2]. Be ready to explain any paragraph in your own words — that is the point of this file. لو فيه جملة مش فاهمها، ارجعلها هنا قبل المناقشة.

7 What is the difference between FIR and IIR in one line?

FIR: no feedback, finite h(k), always stable, can be exactly linear phase, needs more coefficients. IIR: feedback, infinite h(k), may be unstable, nonlinear phase, fewer coefficients, can be derived from analogue filters.